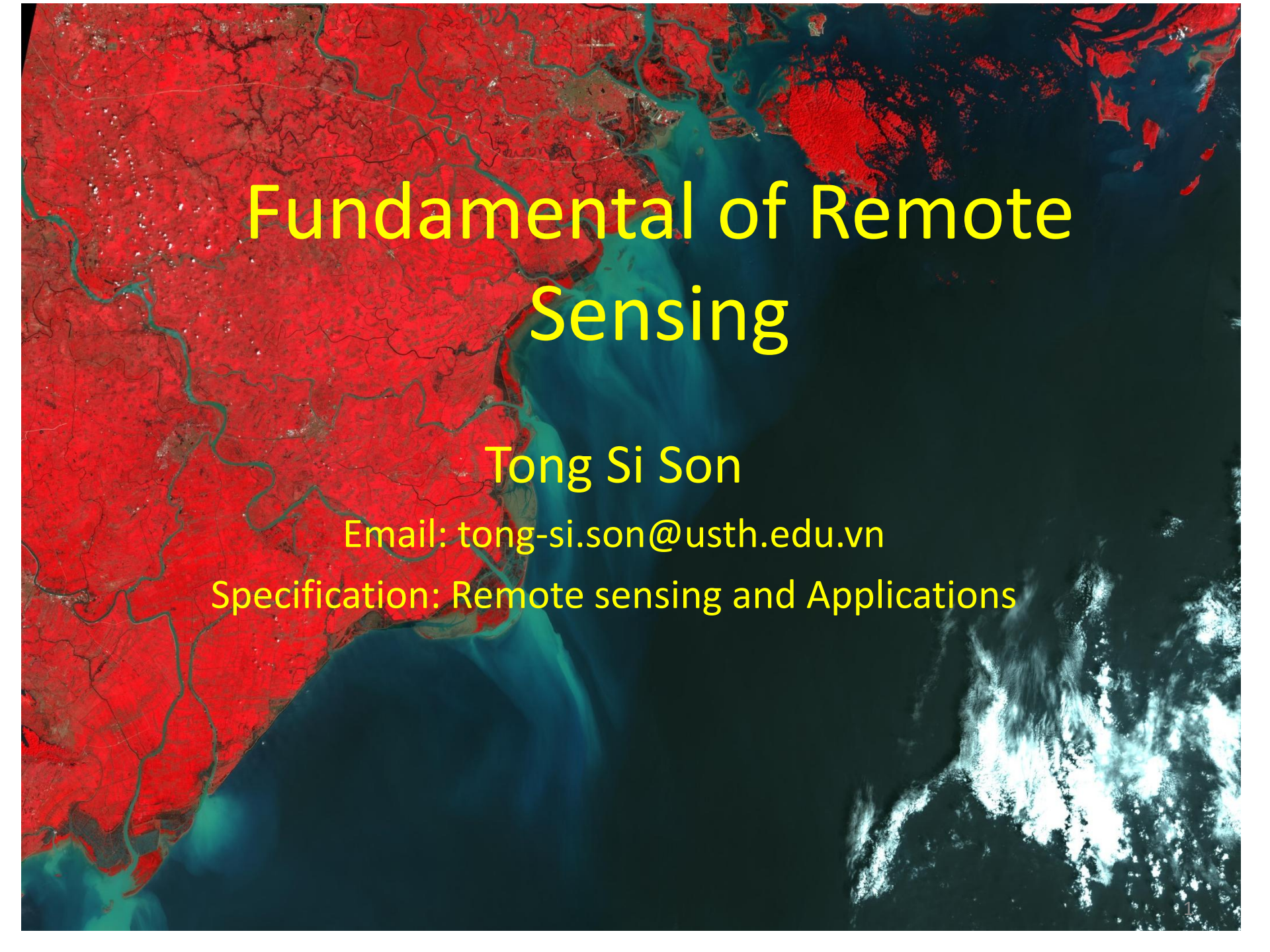


A satellite image of a coastal region. The land is colored red, showing a complex network of rivers and waterways. The surrounding water is dark blue. In the bottom right corner, there is a bright, white, cloud-like feature. The text "Fundamental of Remote Sensing" is overlaid in yellow.

Fundamental of Remote Sensing

Tong Si Son

Email: tong-si.son@usth.edu.vn

Specification: Remote sensing and Applications

Structure Of The Course

- **Time**

36 hours: Lecturer: 21 h, Practice: 15

- **Contents**

1. Overview of Remote sensing
2. Physical basis of Remote sensing
3. Structure of remote sensing systems
- 4. Satellite image characteristics**
5. Satellite image pre-processing
6. Satellite image classification and accuracy assessment

Section 4. Satellite image characteristics

(Material by Prof. Gabriëlle De Lannoy, KU Leuven)

4.1 Spatial resolution

4.2 Spectral resolution

4.3 Radiometric resolution

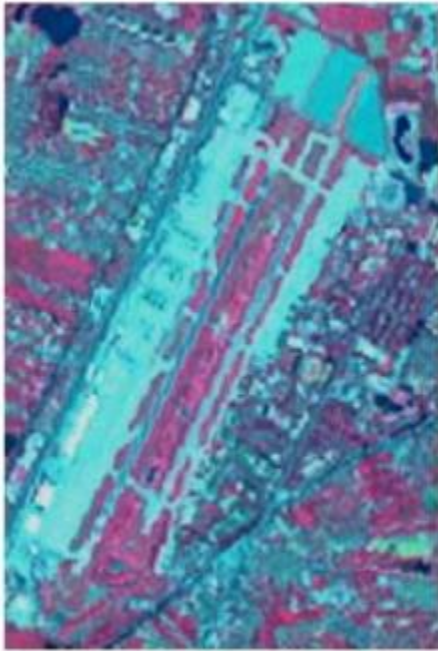
4.4 Temporal resolution

4.5 Resolution tradeoffs

Characteristics of Satellite Images

- **Spatial resolution**, usually measured in pixel size, “is the resolving power of an instrument needed for the discrimination of features and is based on detector size, focal length, and sensor altitude” (NASA, 2013); spatial resolution is also referred to as geometric resolution or IFOV;
- **Spectral resolution**, is the number and location in the electromagnetic spectrum (defined by two wavelengths) of the spectral bands (NASA, 2013) in multispectral sensors, for each band corresponds an image;
- **Radiometric resolution**, usually measured in bits (binary digits), is the range of available brightness values, which in the image correspond to the maximum range of DN's; for example an image with 8 bit resolution has 256 levels of brightness (Richards and Jia, 2006);
- For satellites sensors, there is also the **temporal resolution**, which is the time required for revisiting the same area of the Earth (NASA, 2013).

4.1 Spatial resolution



A) LANDSAT (28.5 m)



B) ALOS AVNIR-2 (10 m)

Comparison of the Detail of 30 x 30 m
Landsat TM Band 3 Data and ALOS
10 x 10 m and IKONOS 1 x 1 m



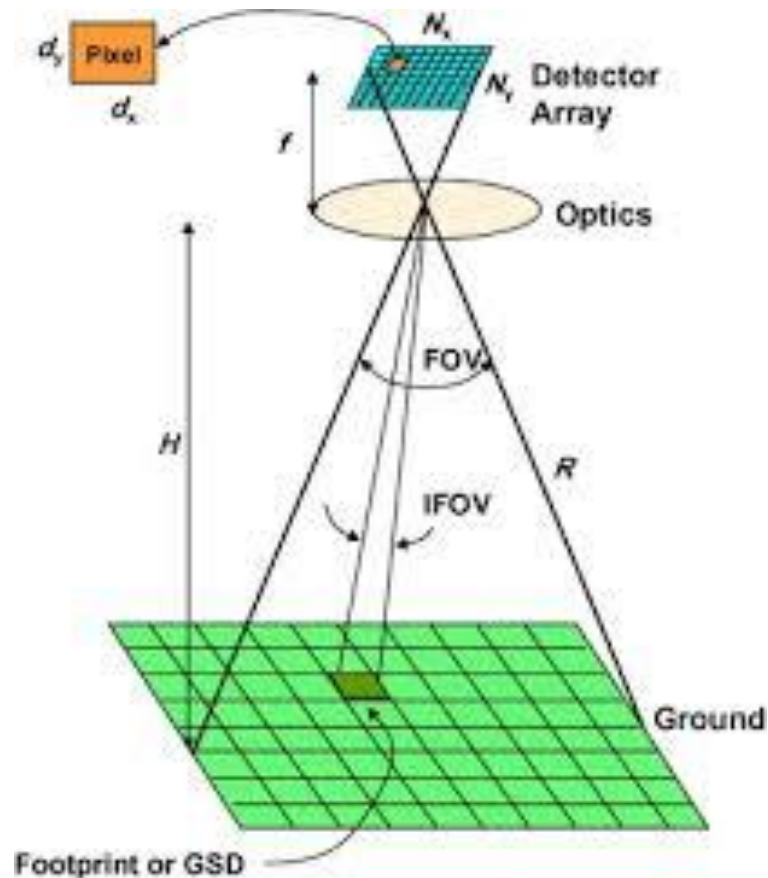
C) Google Earth: IKONOS (1 m)

Spatial resolution

- Spatial Resolution describes how much detail in a photographic image is visible to the human eye.

- Primarily depends on Instantaneous Field of View (IFOV)

- Spatial resolution: depended on geometry, and sensor.



$$\text{Resolution} = \frac{\text{height}}{\text{focal length}} \times \text{detector size}$$

Spatial resolution

Instantaneous Field of View (IFOV): the angle over which a measurement is being made by a detector. Unit: micro radians (μrad).

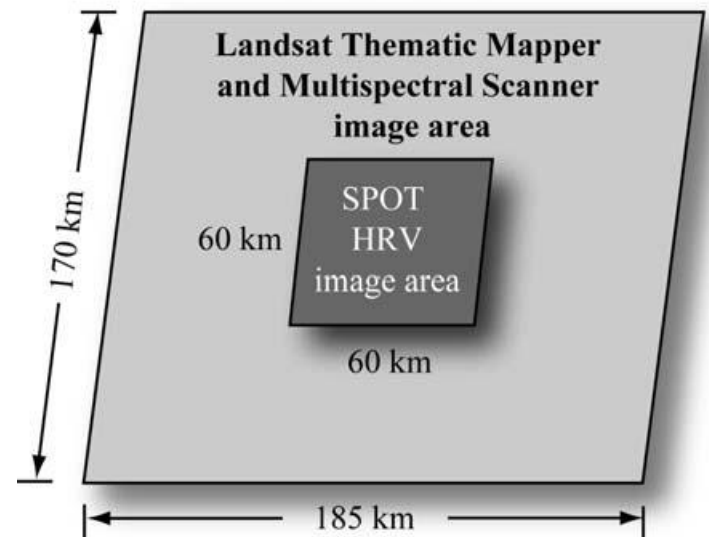
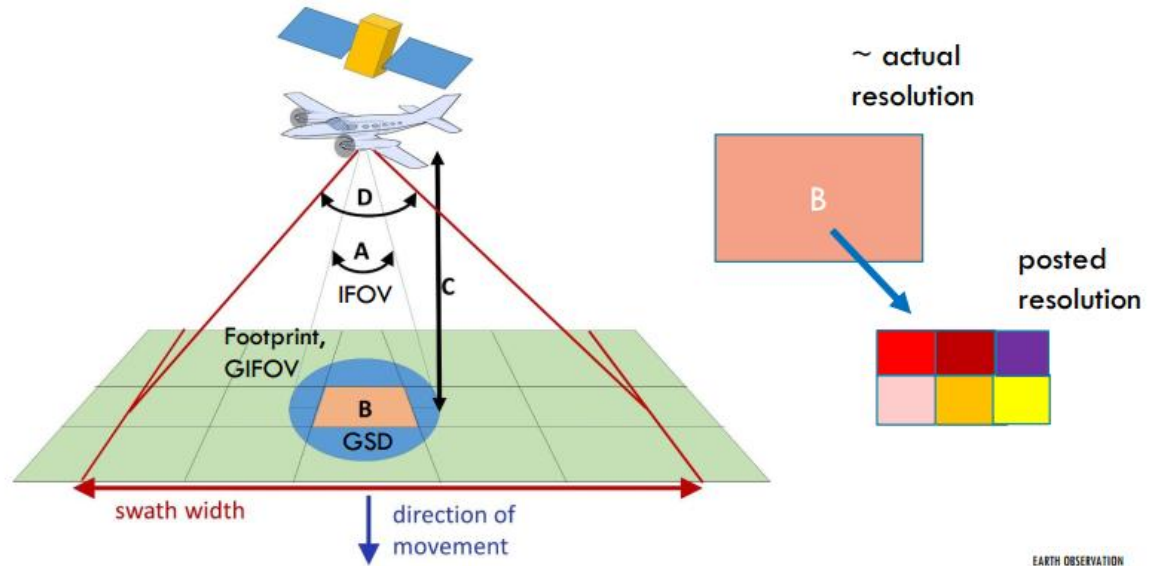
Ground Instantaneous Field of View (GIFOV): ground distance covered by IFOV. Unit: meter (diameter of the circle), dependent on look angle, normal nadir.

Ground Sampling Distance (GSD): the distance between two consecutive pixel centers measured on the ground. Unit: meter. (Normally mentioned as spatial resolution).

Field of view (FOV): Defining the swath width (km)

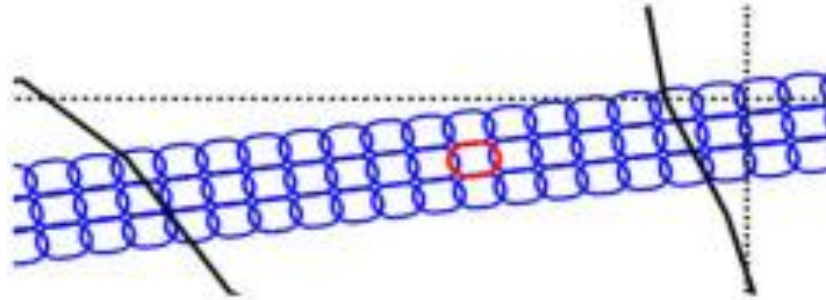
Ex. QuickBird-2: IFOV = $1.37 \mu\text{rad}$
Altitude = $H = 450 \text{ km}$

=>> $\text{GIFOV} = 2 * H * \tan(\text{IFOV}/2) = ?$

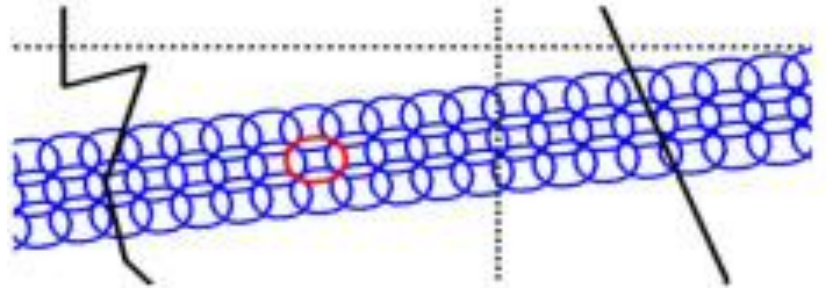


Spatial resolution

Cross-track with little
projection overlap



Cross-track with larger
projection overlap



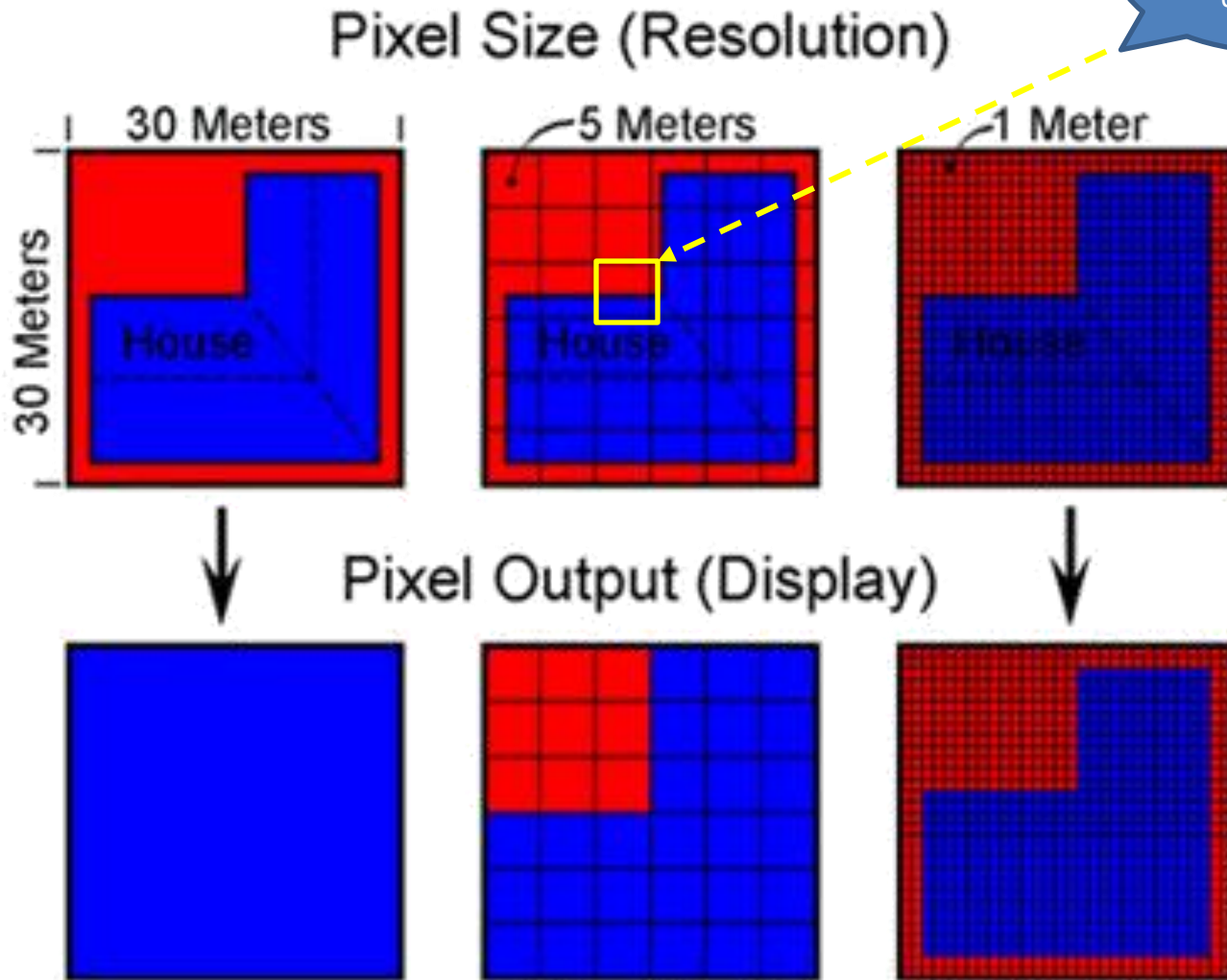
Flight path

Conical scan (fixed off-nadir
angle and rotated in an
azimuthal direction)

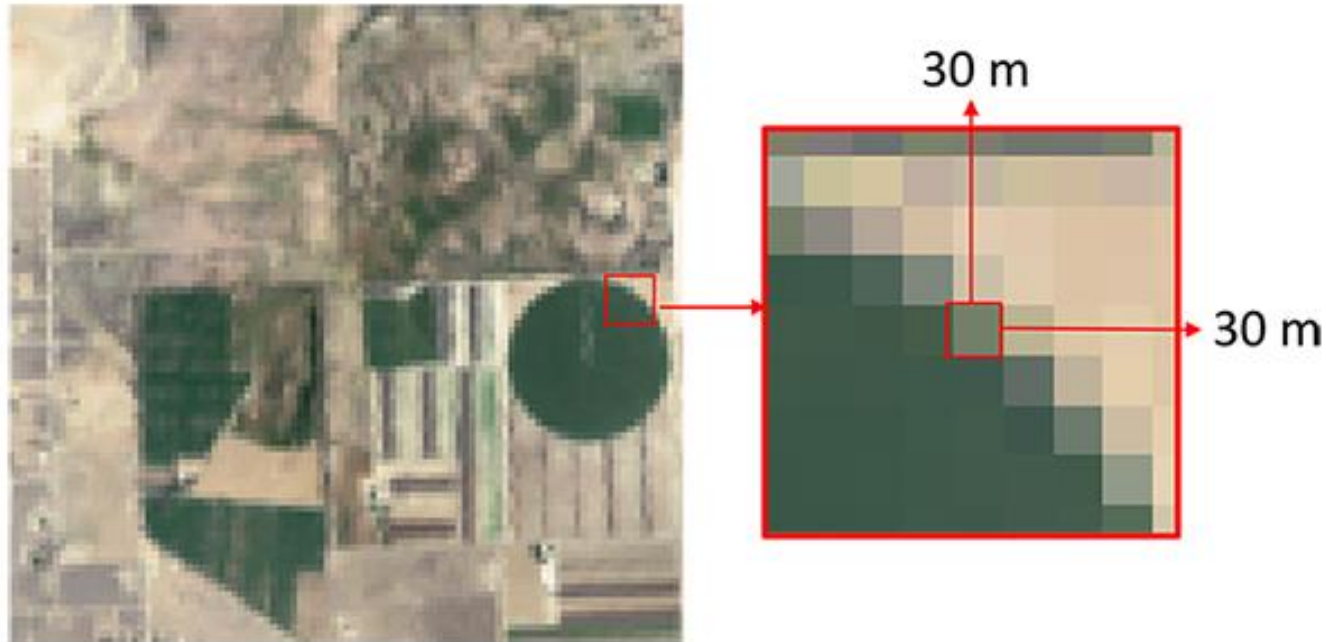


Spatial resolution

How is the DN of this pixel defined?



Spatial resolution



DN, reflectance of overlapped area

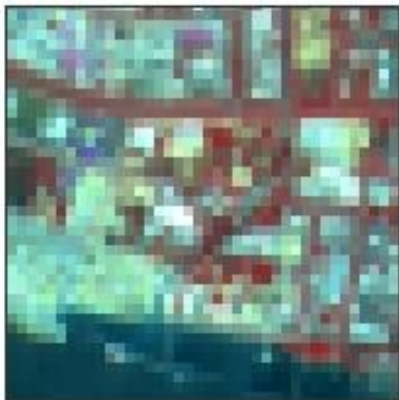
Each Landsat 8 pixel is 30m x 30m or 900m²



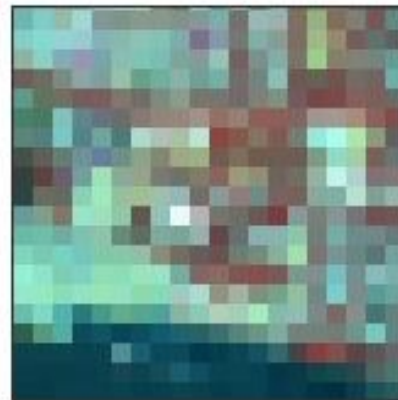
Pixel Size = 10 m
Image Width = 160 pixels, Height = 160 pixels



Pixel Size = 20 m
Image Width = 80 pixels, Height = 80 pixels

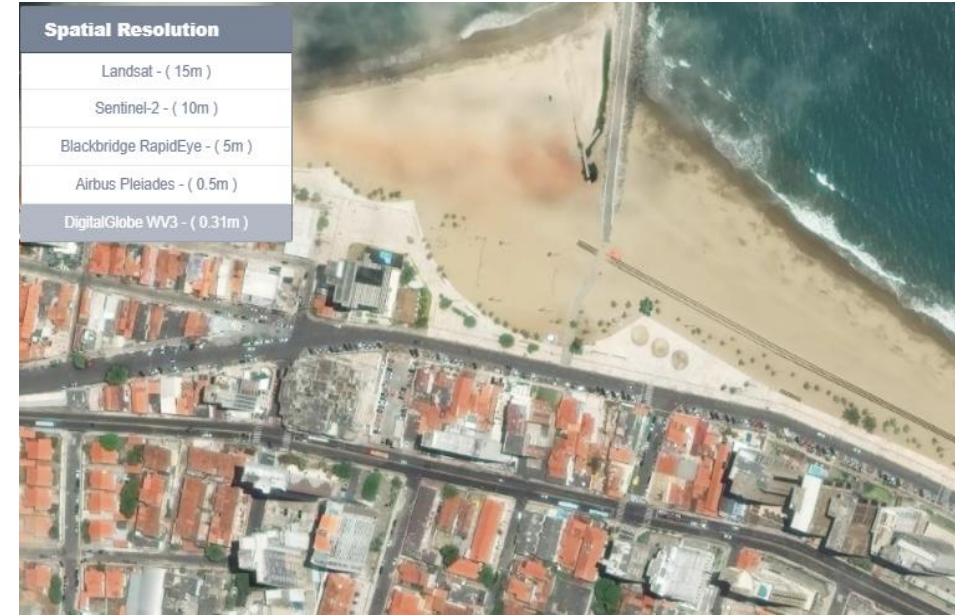
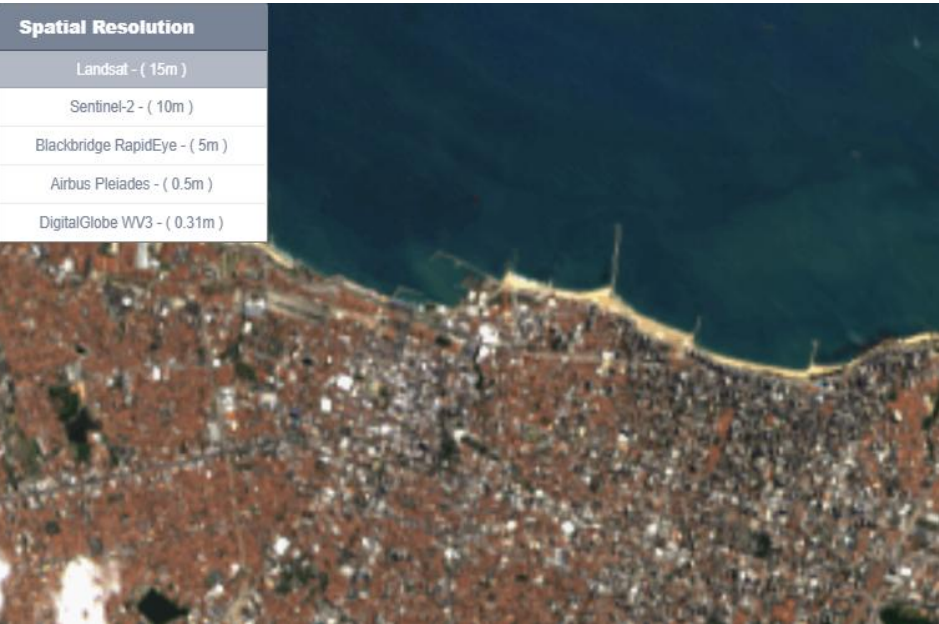


Pixel Size = 40 m
Image Width = 40 pixels, Height = 40 pixels



Pixel Size = 80 m
Image Width = 20 pixels, Height = 20 pixels

An area
represented
in different
resolution



Systems of resolution 1 m or better

Parameter		IKONOS	QuickBird	OrbView-3
Image acquisition mode		Synchronous	Asynchronous	Synchronous
Stereopair acquisition mode		Along-track, re-targeting	Not used	Along-track, re-targeting
Ground sample distance, m	Panchromatic	0.82-1.0	0.61-0.72	1
	Multispectral	3.2-4.0	2.44-2.88	4
Dynamic range, bits per sample		11	11	11
Photogrammetric processing method		RPC	RPC, rigorous	RPC, rigorous
Swath width at nadir, km		11	16.5	8

Systems of resolution 1 m or better

Parameter		KOMPSAT-2	Resurs-DK	EROS B	Cartosat-2
Image acquisition mode		Synchronous	Synchronous	Synchronous and asynchronous	Synchronous and asynchronous
Stereopair acquisition mode		Re-targeting	Re-targeting	Re-targeting	Re-targeting
Ground sample distance, m	Panchromatic	1	1	0.7	1
	Multispectral	4	2-3	-	-
Dynamic range, Bits per pixel		10	10	10	10
Photogrammetric processing method		RPC, rigorous	Rigorous	Rigorous?	RPC?
Swath width at nadir, km		15	28.3 (H=350 km)	7	9.6

Systems of resolution about 2 m

Parameter		SPOT-5	EROS A	Cartosat-1	Formosat-2	ALOS/PRISM
Image acquisition mode		Synchronous	Asynchronous	Synchronous	Synchronous	Synchronous
Stereopair acquisition mode		Cross-track	Along-track, re-targeting	Along-track, two sensors	Cross-track	Along-track, two sensors
Ground sample distance, m	Panchromatic	2.5	1.9	2.5	2	2.5
	Multispectral	2.5	-	-	8	-
Dynamic range, bits per sample		8	10	10	8	8
Photogrammetric processing method		Rigorous	Rigorous	RPC	Rigorous	Rigorous, RPC
Swath width at nadir, km		60	14	26 (stereo)	24	70

Systems of resolution 5 m

Parameter		SPOT-5 / HRG	SPOT-5 / HRS	IRS 1C / PAN IRS 1D / PAN	IRS P6 / LISS IV
Image acquisition mode		Synchronous	Synchronous	Synchronous	Synchronous
Stereopair acquisition mode		Cross-track	Along-track, two sensors	Cross-track	Cross-track
Ground sample distance, m	Panchromatic	5	5×10	5.8	5.8
	Multispectral	5	-	-	5.8
Dynamic range, bits per sample		8	8	6	10 / 7
Photogrammetric processing method		Rigorous	Rigorous	Generic	Generic RPC?
Swath width at nadir, km		60	60	70	23 (Pan) 70 (MS)

Systems of resolution 10-20 m

Parameter		SPOT 1-4	SPOT-5 / HRG MS	Terra/ASTER VNIR	Landsat-7 / ETM+
Image acquisition mode		Synchronous	Synchronous	Synchronous	Whiskbroom
Stereopair acquisition mode		Cross-track	Cross-track	Along-track, two telescopes	-
Ground sample distance, m	Panchromatic	10	-	-	15
	Multispectral	20	10	15	30
Dynamic range, bits per sample		8	8	8	8
Photogrammetric processing method		Rigorous	Rigorous	Rigorous	Generic
Swath width at nadir, km		60	60	60	185

Existing satellite sensors collecting multispectral images with spatial resolution larger than 10 m (2018)

Satellite	Sensor	Spatial Resolution	Revisit Cycle	Operational Period
NOAA	AVHRR	1.1 km	12 h	From 1978
Terra/Aqua	MODIS	Band 1–2: 250 m Band 3–7: 500 m Band 8–36: 1000 m	1 day	From 2000
Terra	ASTER	VNIR: 15 m SWIR: 30 m TIR: 90 m	16 days	From Dec-1999
Landsat	MSS (Landsat 1–3)	79 m	18 days	From 1972 to 1983
	MSS+TM(Landsat5)	VNIR: 30 m TIR: 120 m	16 days	From 1984 to 2013
	ETM+(Landsat-7)	VNIR: 30 m TIR: 60 m	16 days	From 1999
	OLI(Landsat-8)	VNIR: 30 m TIR: 100 m	16 days	From 2013
OrbView-2	Sealifts	1 km	1 day	From 1997
SPOT	HRV(SPOT1–3) VGT(SPOT-4)	20 m 1.15 km	26 days (VGT 1 day)	From 1986
	HRG/HRS/VGT(SPOT-5)	HRG-VNIR: 10 m HRG-SWIR: 20 m		
ENVISAT	MERIS	300 m	35 days	From 2002 to 2012
Sentinel-2	MSI	10 m: (VNIR) B2,3,4,8 20 m: B5,6,7,8A,11,12 60 m: B1,9,10	10 days with one satellite and 5 days with 2 satellites	From Jun-2015
HJ-1A/1B	Multi-spectral sensor	30 m	31 days	From 2009
TH-1	Multi-spectral sensor	10 m	58 days	From Aug-2010
BJ-1	Multi-spectral sensor	32 m		From Oct-2005

CBERS-01/02	Multi-spectral sensor	Multi-spectral-CCD: 19.5 m	26 days	CBERS-01: From 1999 to 2002
		Multi-spectral-IRMSS: 78 m/156 m	26 days	CBERS-02: From Oct-2003
ZY-1 02B	Multi-spectral sensor	20 m	26 days	From Sep-2007
ZY-1 02C	Multi-spectral sensor	10 m	55 days	From Apr-2012
SJ-9A	Multi-spectral sensor	10 m	69 days	From Oct-2012
ALOS	AVNIR-2	10 m	46 days	From 2006 to 2011
ADEOS	Multi-spectral sensor	700 m	41 days	From 1996 to 1997
JERS-1	OPS	18 m	44 days	From 1992 to 1998
IRS-1A/1B	LISS-I/LISS-II/LISS-A/LISS-B	72.5 m(LISS-I); 36 m(LISS-II)	22 days	From 1988
IRS-1C/1D	LISS-III	23.5 m; 70 m	24 days	From 1995 to 2010
IRS-P3	WiFS/MOS	188 m(WiFS); 1500 m/520 m/550 m(MOS)	5 days	From 1996 to 2004
IRS-P6	LISS-IV/LISS-III/AWiFS	5.8 m(LISS-IV)/23.5 m (LISS-III)/70 m(AWiFS)	24 days	From 2003
THEOS	Multi-spectral sensor	15 m	26 days	From Oct-2008

(Zhu et al., 2018)

Main information requirements derived from satellite data

Satellite Sensor	Main information required for treaties that can be derived from satellite data	Related Treaties
Coarse Spatial Resolutoin (1 km – 16 km)	Status & change of forest ecosystems	CBD
	Status & change of habitat types	CBD, Ramsar, WHC
	Habitat fragmentation/destruction	CBD, CITES, CMS, Ramsar, WHC
	Land Cover	CBD, Ramsar
	Forest Fire	CBD, CITES, CMS, Ramsar, WHC
Medium Spatial Resolution (30 m – 500 m)	Status & change of forest ecosystems	CBD
	Status & change of habitat types	CBD, Ramsar, WHC
	Monitoring conservation & protection status	CBD, CITES, CMS, Ramsar
	Habitat range and distribution	CBD, CITES, CMS, Ramsar, WHC
	Habitat fragmentation/destruction	CBD, CITES, CMS, Ramsar, WHC
	Mapping invasive/exotic species	CBD, CMS, Ramsar
	Land Use/land Cover	CBD, Ramsar
	Evaluating the presence of corridors	CBD
	Species distribution	CBD, CITES, CMS, Ramsar
	Forest Fire	CBD, CITES, CMS, Ramsar, WHC
High Spatial Resolution (1 m – 30 m)	Status & change of forest ecosystems	CBD
	Status & change of habitat types	CBD, Ramsar, WHC
	Monitoring conservation & protection effectiveness	CBD, CITES, CMS, Ramsar
	Habitat range and distribution	CBD, CITES, CMS, Ramsar, WHC
	Critical habitat delineation	CBD, CITES, CMS, Ramsar, WHC
	Migration routes	CMS, Ramsar
	Habitat fragmentation/destruction	CBD, CITES, CMS, Ramsar, WHC
	Mapping invasive/exotic species	CBD, CMS, Ramsar
	Evaluating the presence of corridors	CBD
	Species distribution	CBD, CITES, CMS, Ramsar
	Biomass	CBD
	Land Use/Land Cover	CBD, Ramsar
	Leaf Area Index	CBD
	Net Primary Production	CBD, Ramsar
	Road Density	CBD, CITES, Ramsar
	Forest Fire	CBD, CITES, CMS, Ramsar, WHC
	Logging	CBD, CITES, CMS, Ramsar, WHC

Convention on Biological Diversity (CBD)

Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES)

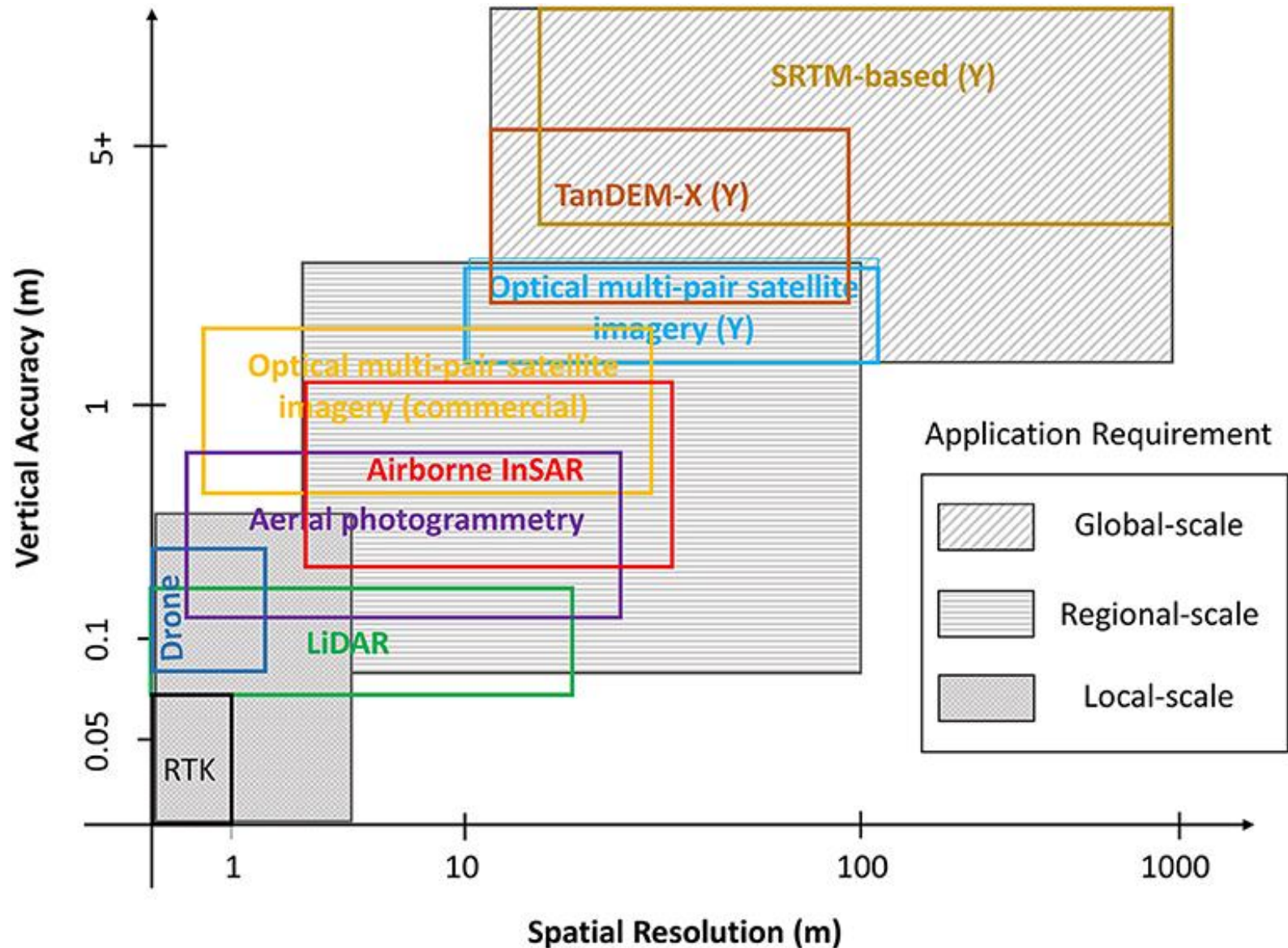
Convention on Migratory Species of Wild Animals (CMS)

Convention on Wetlands of International Importance especially as Waterfowl Habitat (Ramsar)

Convention Concerning the Protection of the World Cultural and Natural Heritage (WHC)

([Sherbinin](#) and [Chandra](#), 2000)

Available DEM technology in context of requirements in spatial resolution and vertical accuracies for different scales in application



(Schumann and Bates, 2018)

Relationship between digitizer instantaneous field-of-view measured in dots per inch or micrometers, and the pixel ground resolution at various scales of photography.

Digitizer Detector IFOV		Pixel Ground Resolution at Various Scales of Photography (m)					
Dots per inch	Micrometers	1:40,000	1:20,000	1:9,600	1:4,800	1:2,400	1:1,200
100	254.00	10.16	5.08	2.44	1.22	0.61	0.30
200	127.00	5.08	2.54	1.22	0.61	0.30	0.15
300	84.67	3.39	1.69	0.81	0.41	0.20	0.10
400	63.50	2.54	1.27	0.61	0.30	0.15	0.08
500	50.80	2.03	1.02	0.49	0.24	0.12	0.06
600	42.34	1.69	0.85	0.41	0.20	0.10	0.05
700	36.29	1.45	0.73	0.35	0.17	0.09	0.04
800	31.75	1.27	0.64	0.30	0.15	0.08	0.04
900	28.23	1.13	0.56	0.27	0.14	0.07	0.03
1000	25.40	1.02	0.51	0.24	0.12	0.06	0.03
1200	21.17	0.85	0.42	0.20	0.10	0.05	0.03
1500	16.94	0.67	0.34	0.16	0.08	0.04	0.02
2000	12.70	0.51	0.25	0.12	0.06	0.03	0.02
3000	8.47	0.33	0.17	0.08	0.04	0.02	0.01
4000	6.35	0.25	0.13	0.06	0.03	0.02	0.008
Useful Scanning Conversions DPI = dots per inch; μm = micrometers; I = inches; M = meters From DPI to micrometers: $\mu\text{m} = (2.54 / \text{DPI})10,000$ From micrometers to DPI: $\text{DPI} = (2.54 / \mu\text{m})10,000$ From inches to meters: $M = I \times 0.0254$ From meters to inches: $I = M \times 39.37$							
Computation of Pixel Ground Resolution PM = pixel size in meters; PF = pixel size in feet; S = photo scale Using DPI: $\text{PM} = (S/\text{DPI})/39.37$ $\text{PF} = (S/\text{DPI})/12$ Using micrometers: $\text{PM} = (S \times \mu\text{m}) 0.000001$ $\text{PF} = (S \times \mu\text{m}) 0.00000328$ For example, if a 1:6,000 scale aerial photograph is scanned at 500 dpi, the pixel size will be $(6000/500)/39.37 = 0.3048$ meters per pixel, or $(6000/500)/12 = 1.00$ foot per pixel. If a 1:9,600 scale aerial photograph is scanned at 50.8 μm , the pixel size will be $(9,600 \times 50.8)(0.000001) = 0.49$ meters, or $(9,600 \times 50.8)(0.00000328) = 1.6$ feet per pixel.							

Spatial resolution

26/8/2018 (2.5 m)



26/8/2018 (30 m)



How many classes can be defined in each image ?

Spatial Resolution Trade-offs

- Data volume
- Signal to Noise Ratio
- “Salt and Pepper”
- Cost

4.2 Spectral Resolution

Can be described two ways, but they usually go hand in hand.

- How many spectral “bands” an instrument records
- How “wide” each band is (the range of wavelengths covered by a single band)

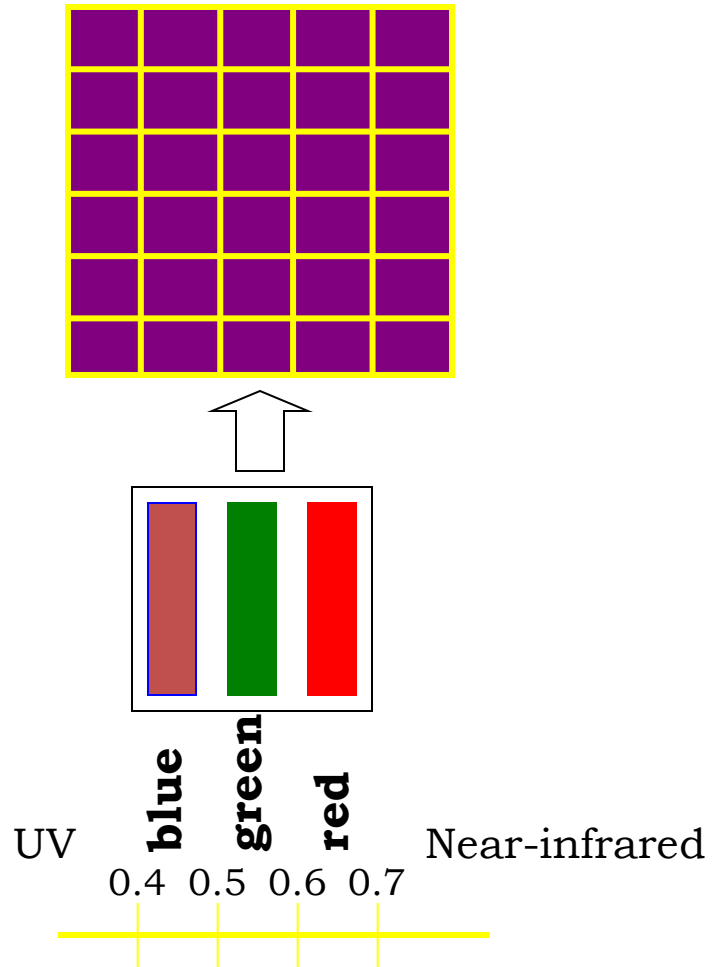
Spectral resolution

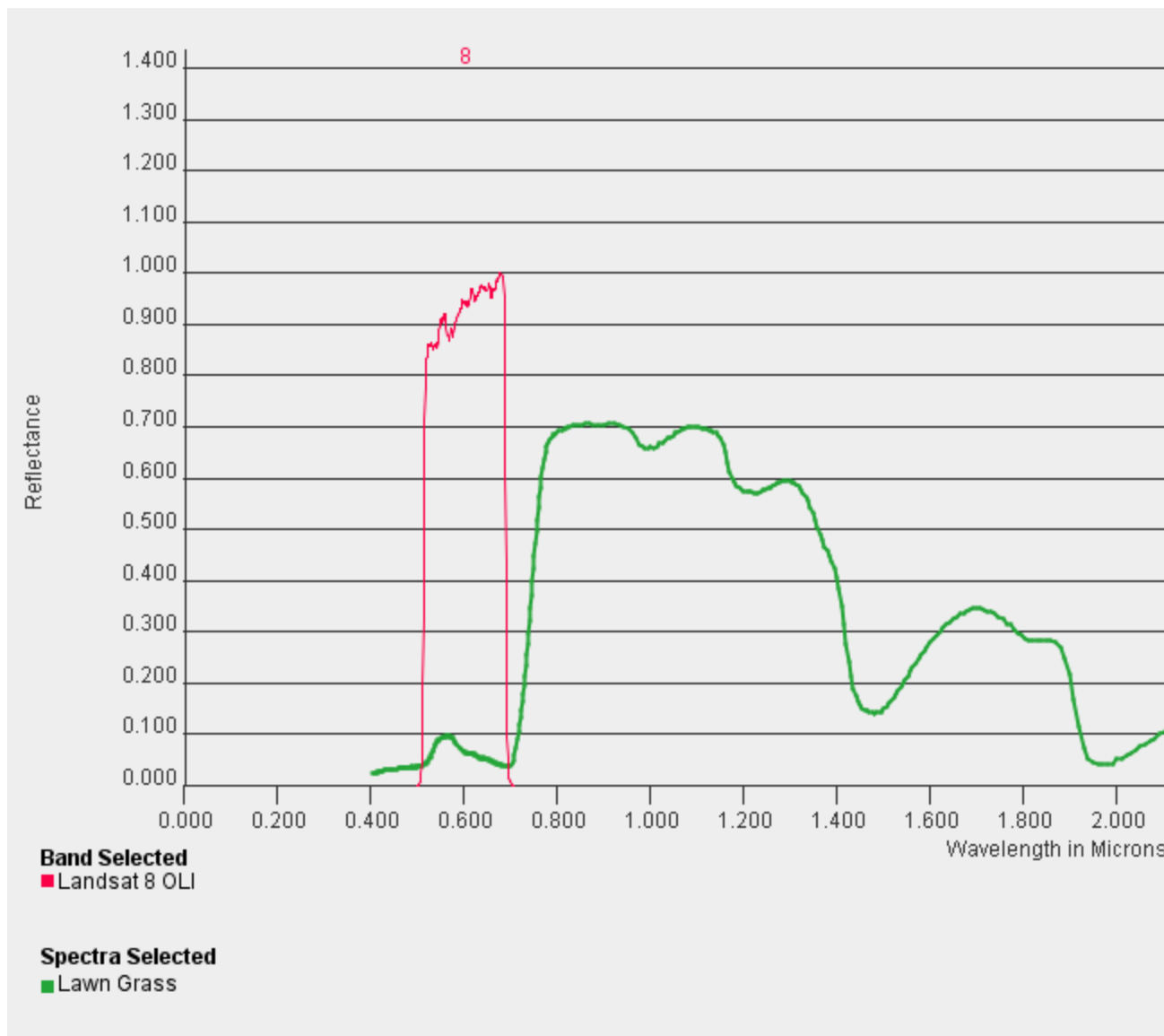
- Related to the measured range of EMR
- Wide range - coarser resolution
- Narrow range - finer resolution

Case 1

- Measure the EMR across a wide range
- *E.g.*, a single *panchromatic band* covering the entire visible portion of the spectrum
- Assigns a single DN representing all visible light energy hitting the sensor
- Analogous to black and white (panchromatic) film

Case 1



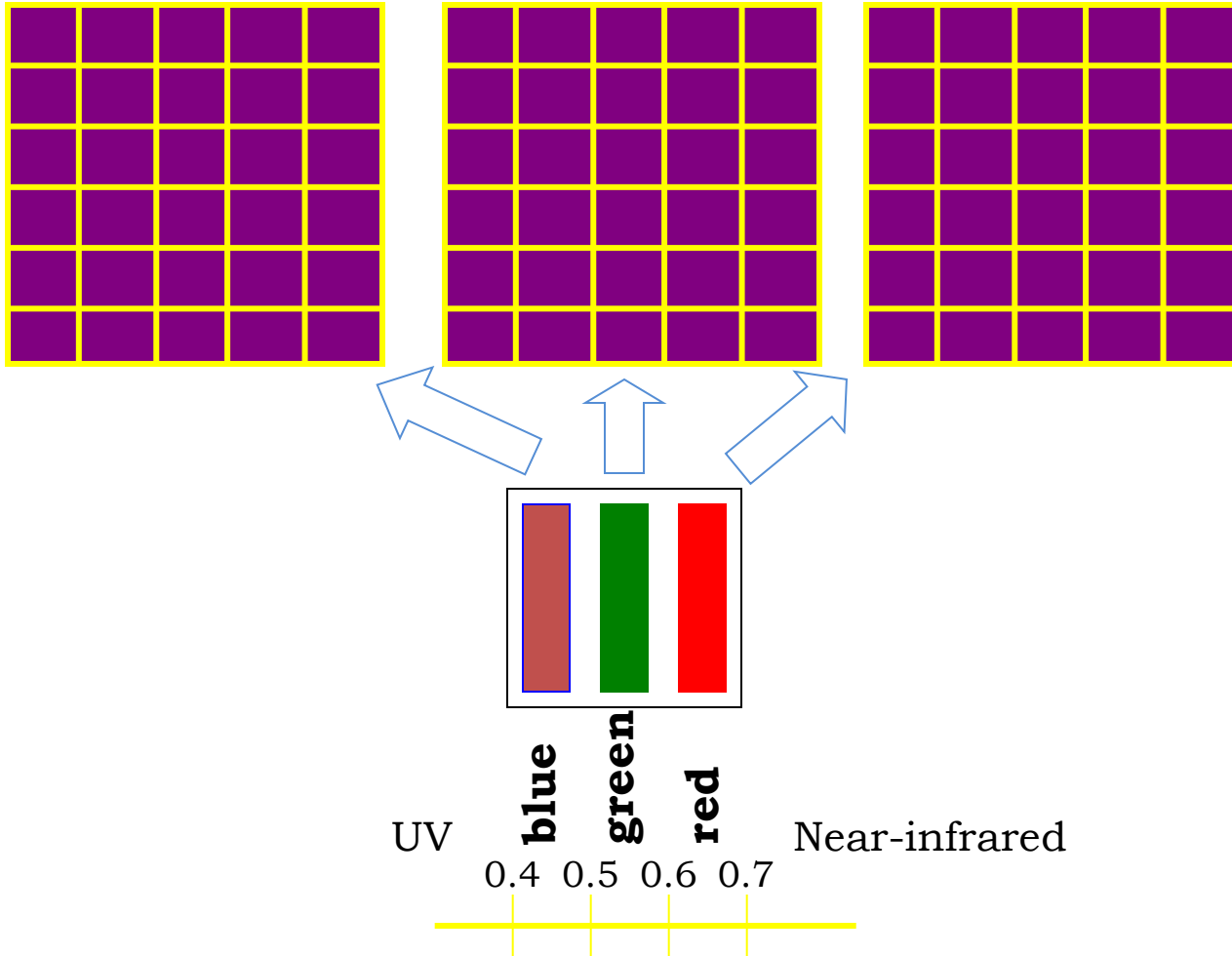


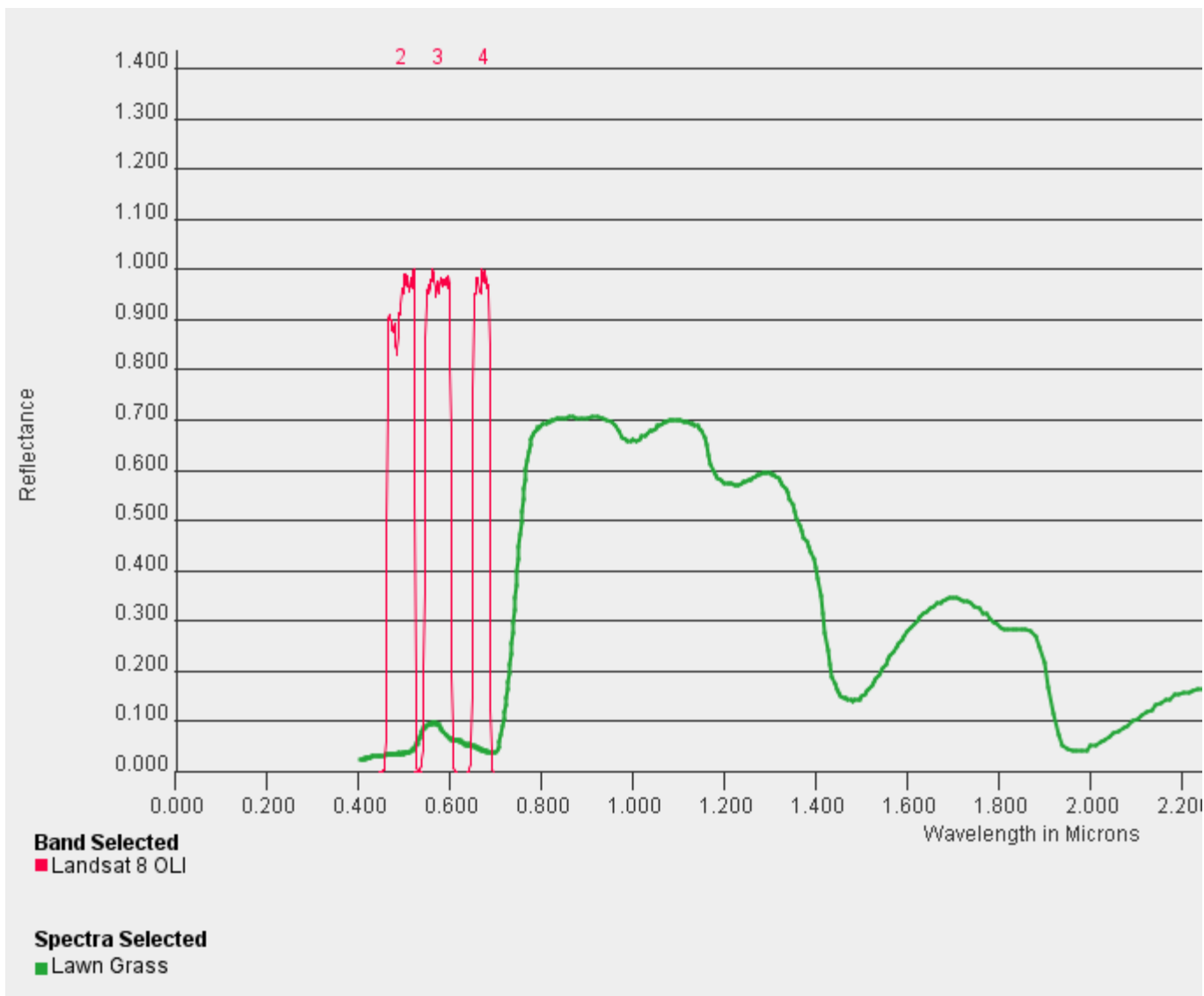
From [USGS Spectral Characteristics Viewer](#)

Case 2

- Measure EMR across narrower ranges
- *E.g.*, Separate bands for blue, green and red parts of the spectrum
- Assign a DN for *each* of these wavelength ranges to create *3 bands*

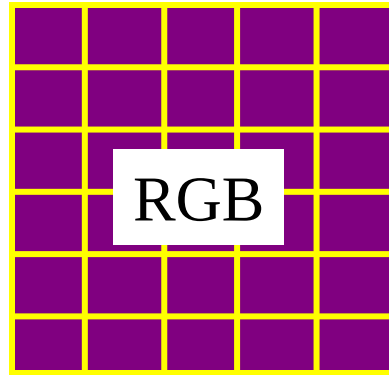
Case 2



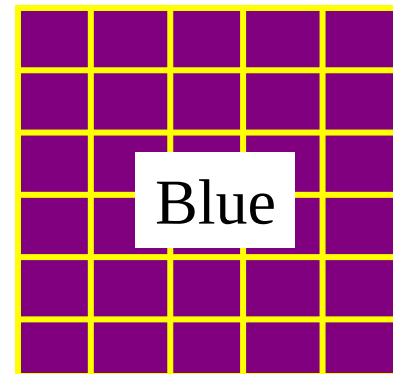
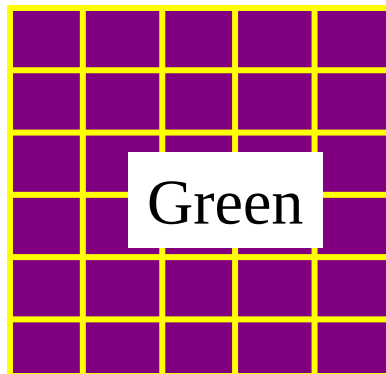
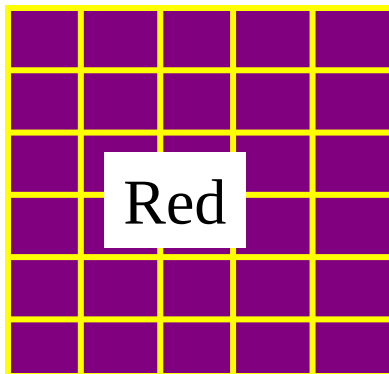


From [USGS Spectral Characteristics Viewer](#)

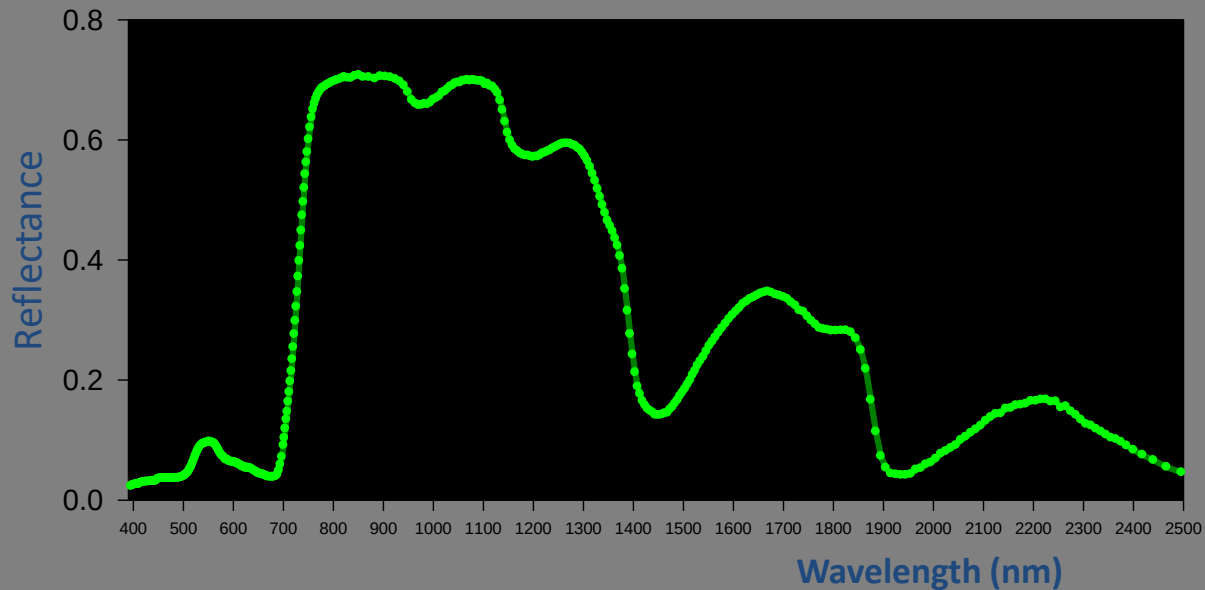
Coarser (lower) Spectral Resolution



Finer (higher) Spectral Resolution

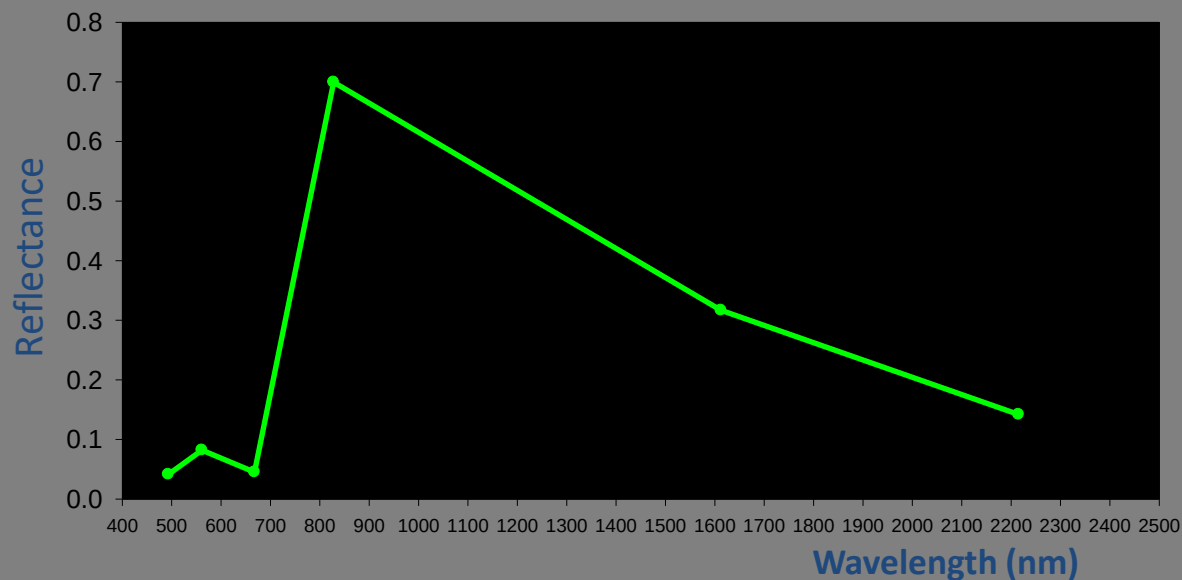


High Spectral Resolution

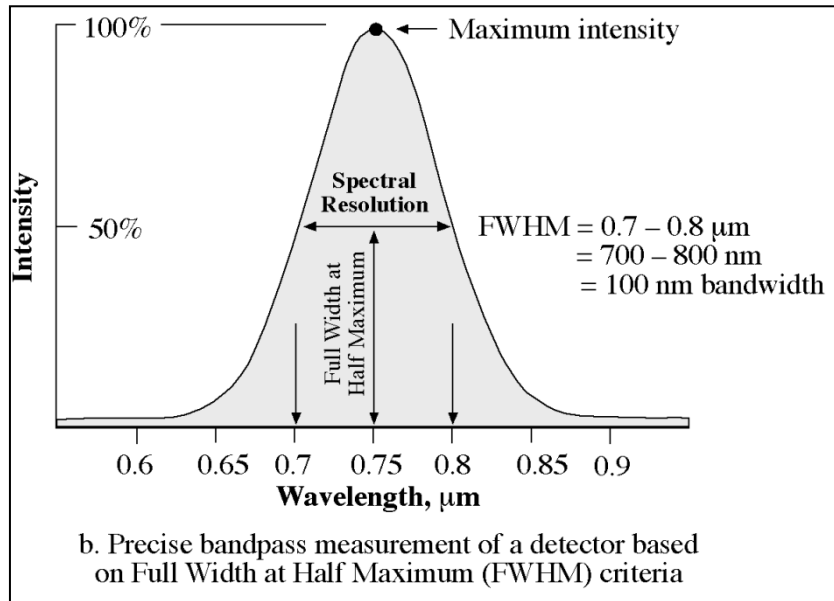


Spectral reflectance curve for green leaf using 224 bands (high spectral resolution)

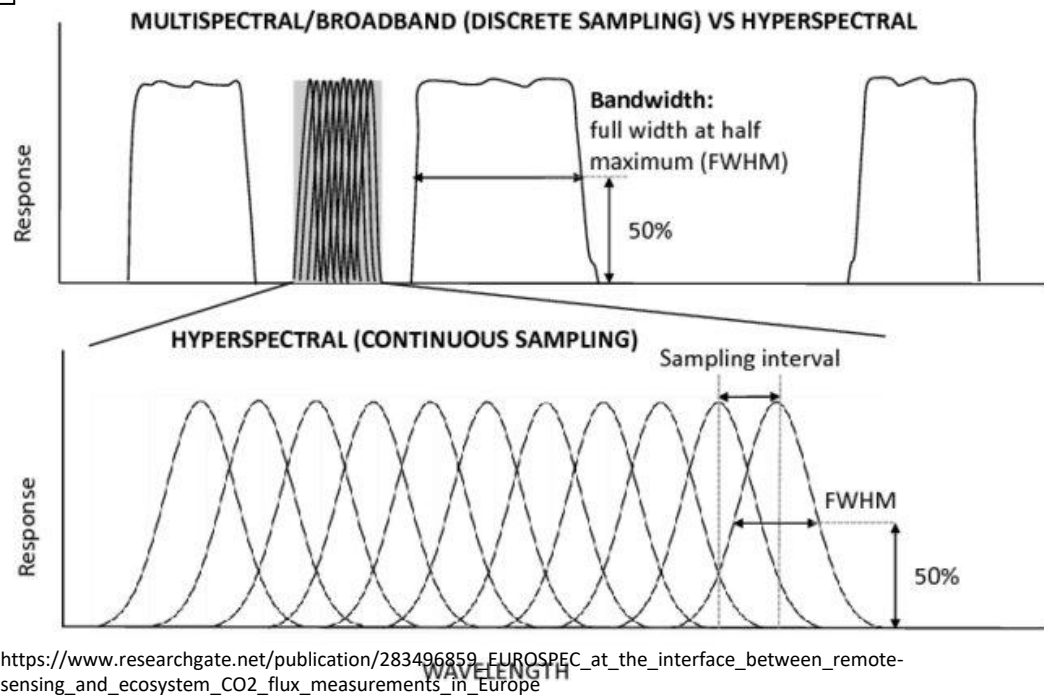
Low Spectral Resolution



Spectral reflectance curve for green leaf using 6 bands (lower spectral resolution)



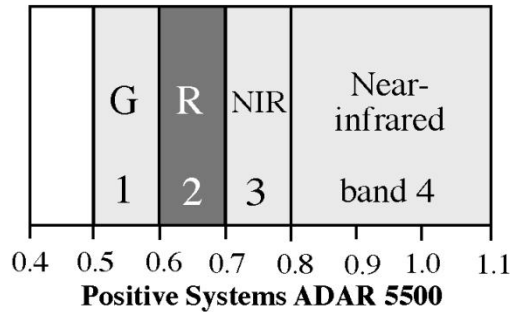
Full Width at Half maximum (FWHM) criteria:
 Band (ex. 0.7-0.8) record light energy in a range of wavelength 0.6-0.9,
 Band width: 100 nm (0.7-0.8 micrometer) $\Rightarrow$ adjacent bands have an overlaps wavelength area.



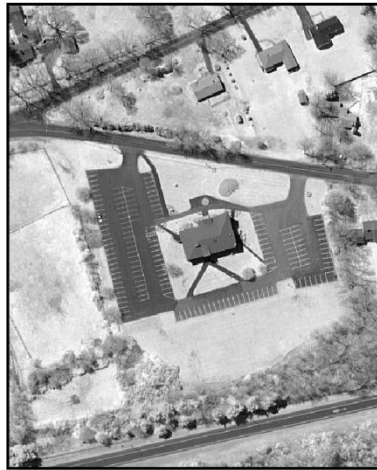
Spectral resolution

Number of bands and The width of bands

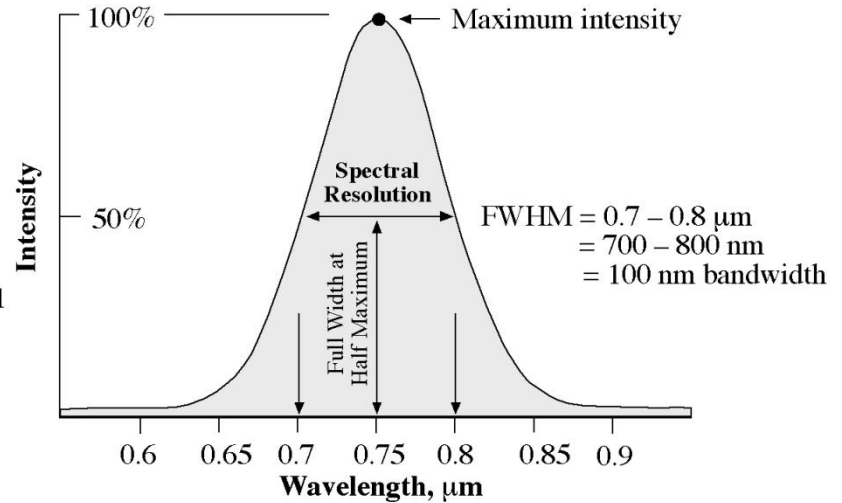
Landsat Multispectral Scanner (MSS)



a. Nominal spectral resolution of the Landsat Multispectral Scanner and Positive Systems ADAR 5500 digital frame camera.



c. Single band of ADAR 5500 data



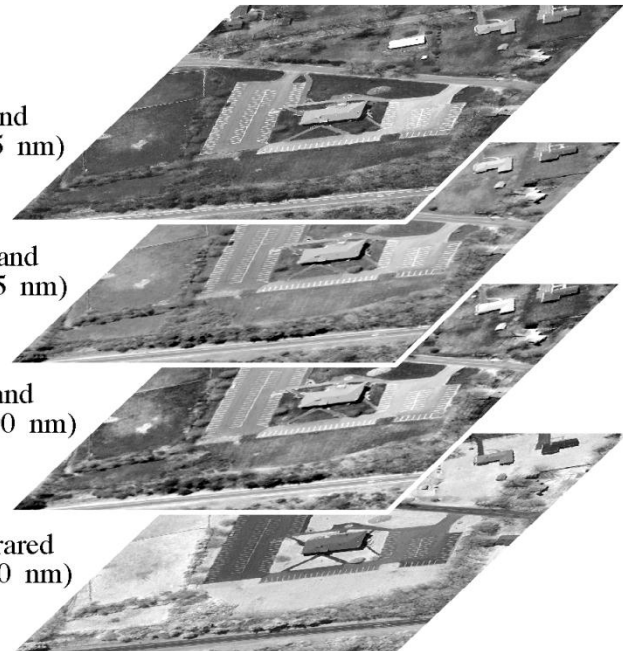
b. Precise bandpass measurement of a detector based on Full Width at Half Maximum (FWHM) criteria

blue band
(450 - 515 nm)

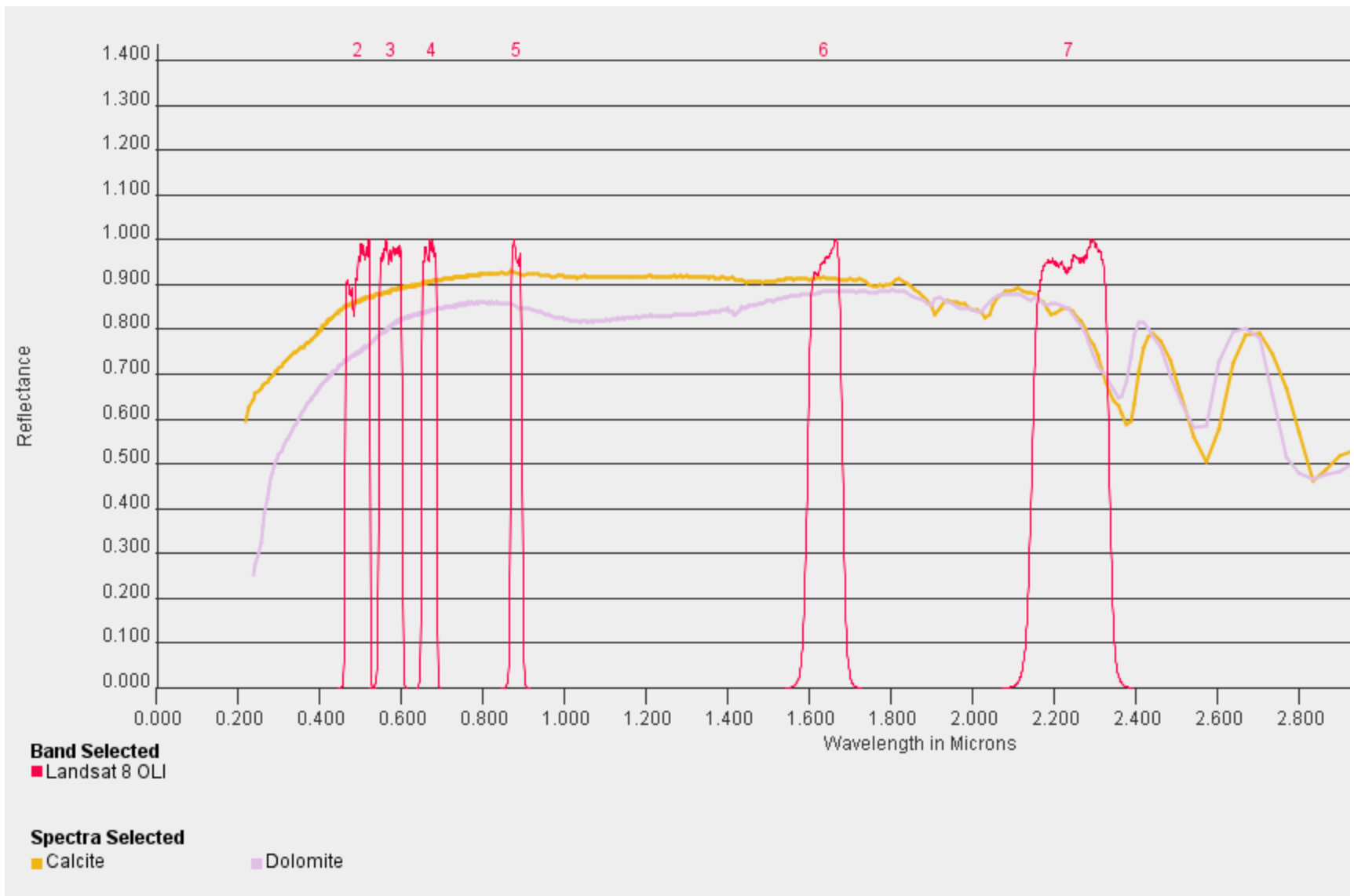
green band
(525 - 605 nm)

red band
(640 - 690 nm)

near-infrared
(750 - 900 nm)

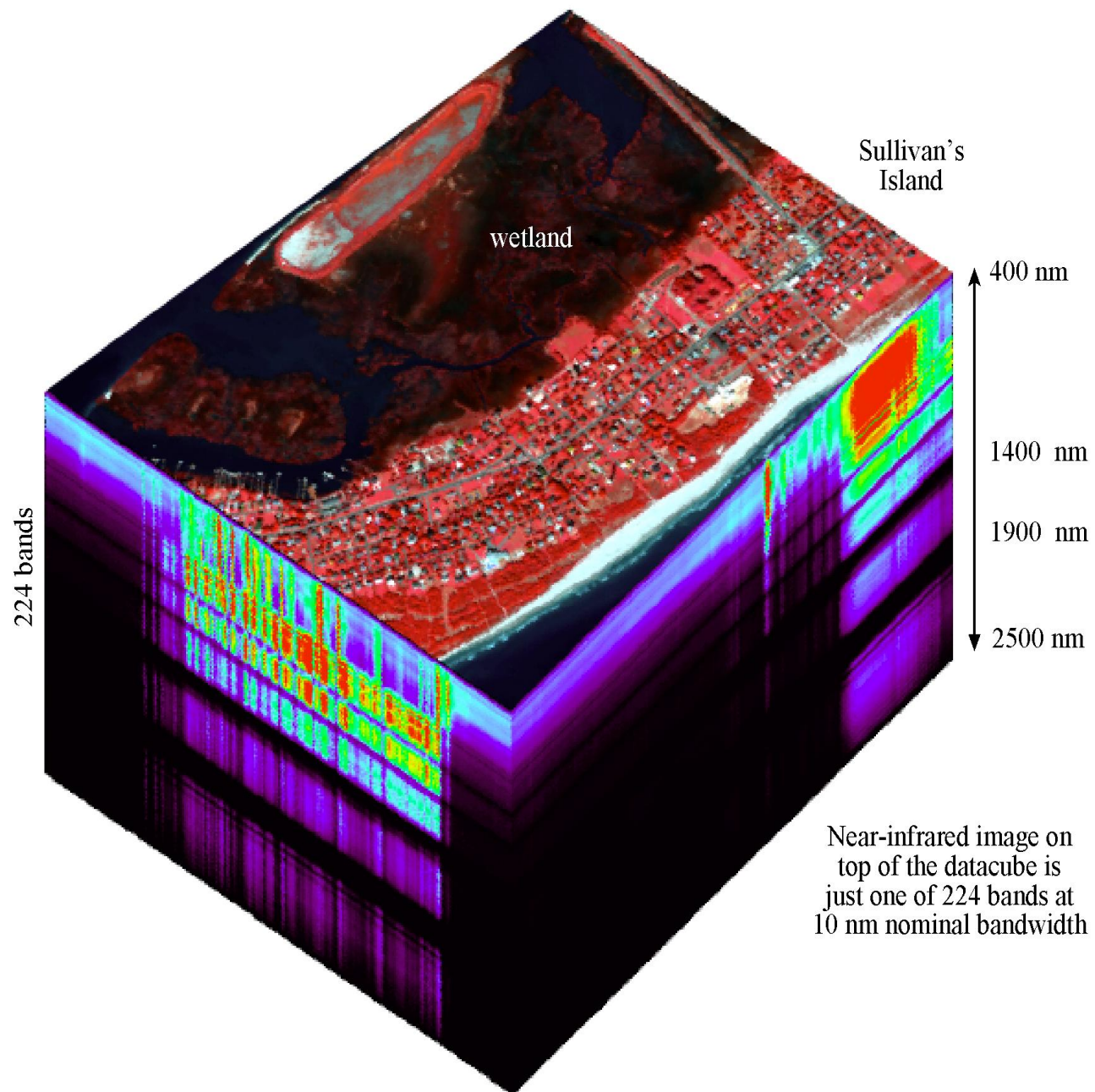


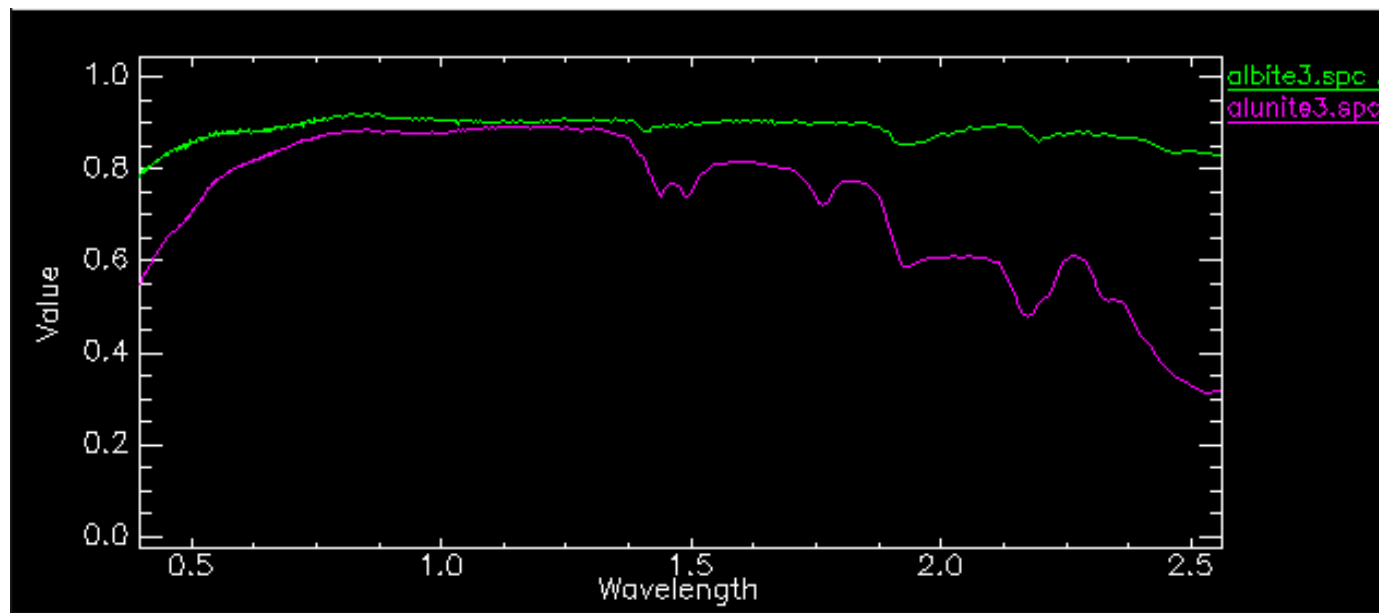
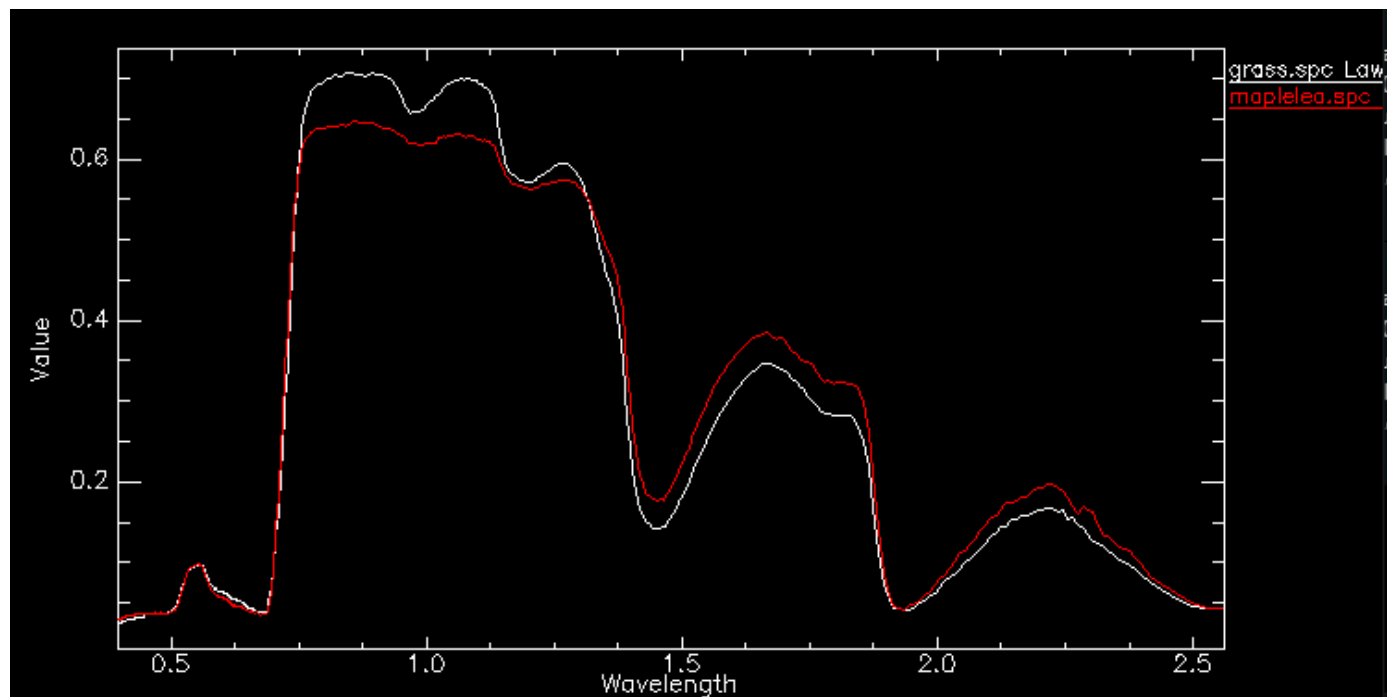
d. Multispectral remote sensing



Could you distinguish Dolomite from Calcite using Landsat 8 spectral data?

Spectral resolution





Spectral resolution

Landsat 8 Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS) Launched February 11, 2013	Bands	Wavelength (micrometers)	Resolution (meters)
	Band 1 - Coastal aerosol	0.43 - 0.45	30
	Band 2 - Blue	0.45 - 0.51	30
	Band 3 - Green	0.53 - 0.59	30
	Band 4 - Red	0.64 - 0.67	30
	Band 5 - Near Infrared (NIR)	0.85 - 0.88	30
	Band 6 - SWIR 1	1.57 - 1.65	30
	Band 7 - SWIR 2	2.11 - 2.29	30
	Band 8 - Panchromatic	0.50 - 0.68	15
	Band 9 - Cirrus	1.36 - 1.38	30
	Band 10 - Thermal Infrared (TIRS) 1	10.60 - 11.19	100
	Band 11 - Thermal Infrared (TIRS) 2	11.50 - 12.51	100

Satellite	Band no	Spectral region	Spatial resolution (m)	Wavelengths (nm)
Sentinel-2	2	Blue	10 m × 10 m	458–523
Sentinel-2	3	Green	10 m × 10 m	543–578
Sentinel-2	4	Red	10 m × 10 m	650–680
Sentinel-2	8	NIR	10 m × 10 m	785–900
Sentinel-2	5	Red edge/NIR	20 m × 20 m	698–713
Sentinel-2	6	Red edge/NIR	20 m × 20 m	733–748
Sentinel-2	7	Red edge/NIR	20 m × 20 m	773–793
Sentinel-2	8a	Red edge/NIR	20 m × 20 m	855–875
DMC	1	Green	22 m × 22 m	520–600
DMC	2	Red	22 m × 22 m	630–690
DMC	3	NIR	22 m × 22 m	770–900

Spectral resolution

Satellite sensor	Launch date	Spectral bandwidths	Spatial resolution (m)		
Landsat ETM+	1999	Blue: 450–515 nm	30		
		Green: 525–605 nm	30		
		Red: 630–690 nm	30		
		NIR: 750–900 nm	30		
		MIR: 1550–1750 nm	30		
		TIR: 10400–12500 nm	60		
		SWIR: 2090–2350 nm	30		
		PAN: 520–900 nm	15		
Hyperion ^a	1999	Hyperspectral 400–2500 nm 220 bands 10 nm resolution	30		
		CHRIS ^a	2001	Hyperspectral 400–1050 nm 19–63 bands 1.3–12 nm resolution	18–36
				SPOT-5	2002
Green: 610–680 nm	10				
Red: 790–890 nm	10				
SWIR: 1580–1750 nm	20				
PAN: 510–730 nm	5				
IKONOS	1999	Blue: 445–516 nm	4		
		Green: 506–595 nm	4		
		Red: 632–698 nm	4		
		NIR: 757–853 nm	4		
		PAN: 450–900 nm	1		
NIR, near infrared; MIR, mid infrared; TIR, thermal infrared; SWIR, short-wave infrared; PAN, panchromatic.					
^a Experimental prototype sensors.					

Spectral resolution

27/12/2010 (Panchromatic)



19/01/2015 (Natural color composite)



How many classes can be defined in each image ?

4.3 Radiometric Resolution

- How finely does the satellite divide up the radiance it receives in each band?

How much light does it take to change the DN from one number to the next?

- Usually expressed as number of bits used to store the maximum possible DN value
 - 8 bits = $2^8 = 256$ levels (usually 0 to 255)
 - 16 bits = $2^{16} = 65,536$ levels (0 to 65,535)

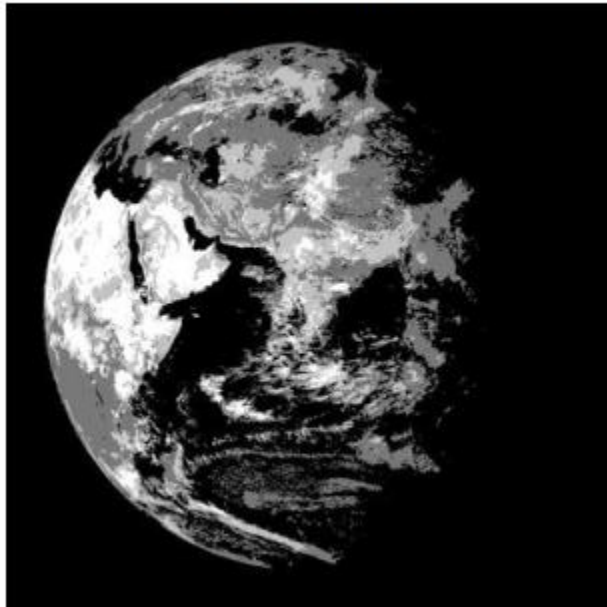
$2^6 = 64$ levels (6 bit)



$2^2 = 4$ levels (2 bit)



Gray scale 4



Gray scale 16



Gray scale 256



Same size, same spatial resolution but different radiometric resolution



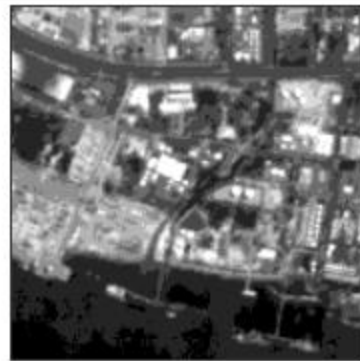
8-bit quantization (256 levels)



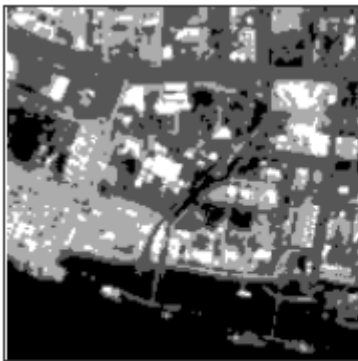
6-bit quantization (64 levels)



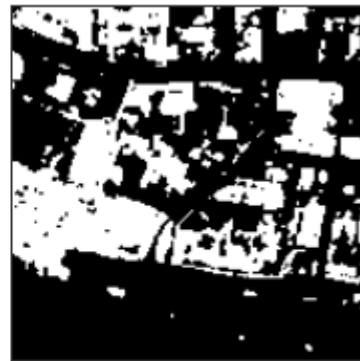
4-bit quantization (16 levels)



3-bit quantization (8 levels)



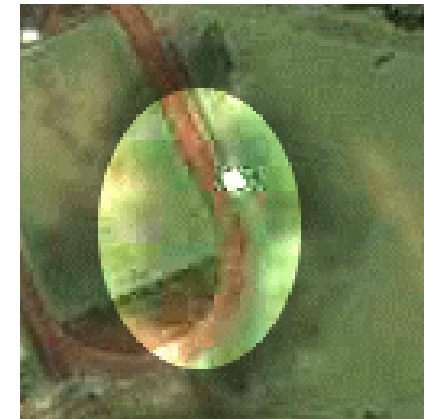
2-bit quantization (4 levels)



1-bit quantization (2 levels)



Part of the running track the **IKONOS** (11-bit) image is under cloud shadow. The high radiometric resolution enables features under shadow to be recovered.



The features under cloud shadow are recovered by applying a simple contrast and brightness enhancement technique.

Radiometric resolution

- 1 bit (0 - 1)
- 8 bit (0 - 255) (older Landsats, many others)
- 16 bit (0 - 65,535) (Landsat 8)
- 32 bit (0 - 4,294,967,295) (uncommon)

For an 8-bit satellite:

DN = 0: No EMR or below some minimum
amount of light (threshold)

DN = 255: Max EMR or above some maximum
amount of light

Radiometric resolution

- 8-bit data (*e.g.*, Landsat 5) (256 values)
 - Everything will be scaled from 0 – 255
 - Subtle details may not be represented
- 16-bit data (*e.g.*, Landsat 8) (65,536 values)
 - Wide range of choices
 - Required storage space will be twice that of 8 bit

Radiometric Radiation Trade Offs

- Data volume

Every 8 bits takes 1 byte to store on a computer.

One 8-bit DN takes 1 byte

One 16-bit DN takes 2 bytes

Etc.

Calculating Image Size

Computer hard drives store data in “boxes” called ***bytes***

1 byte can hold 8 binary (base 2) digits (0s or 1s or some combination of 0s and 1s)

Each “bit” is a single binary digit

An 8-bit number is made of 8 binary digits and fits into 1 byte.

A 9-bit number won't fit in 1 byte and requires 2 bytes.

Group Problem

- If you are using 7-band, 16-bit Landsat 8 data for your 1 km² area, how many bytes are needed to store your data on your computer?

4.4 Temporal resolution

- Time between two subsequent data acquisitions for an area

All of the Landsat satellites have a 16-day return time

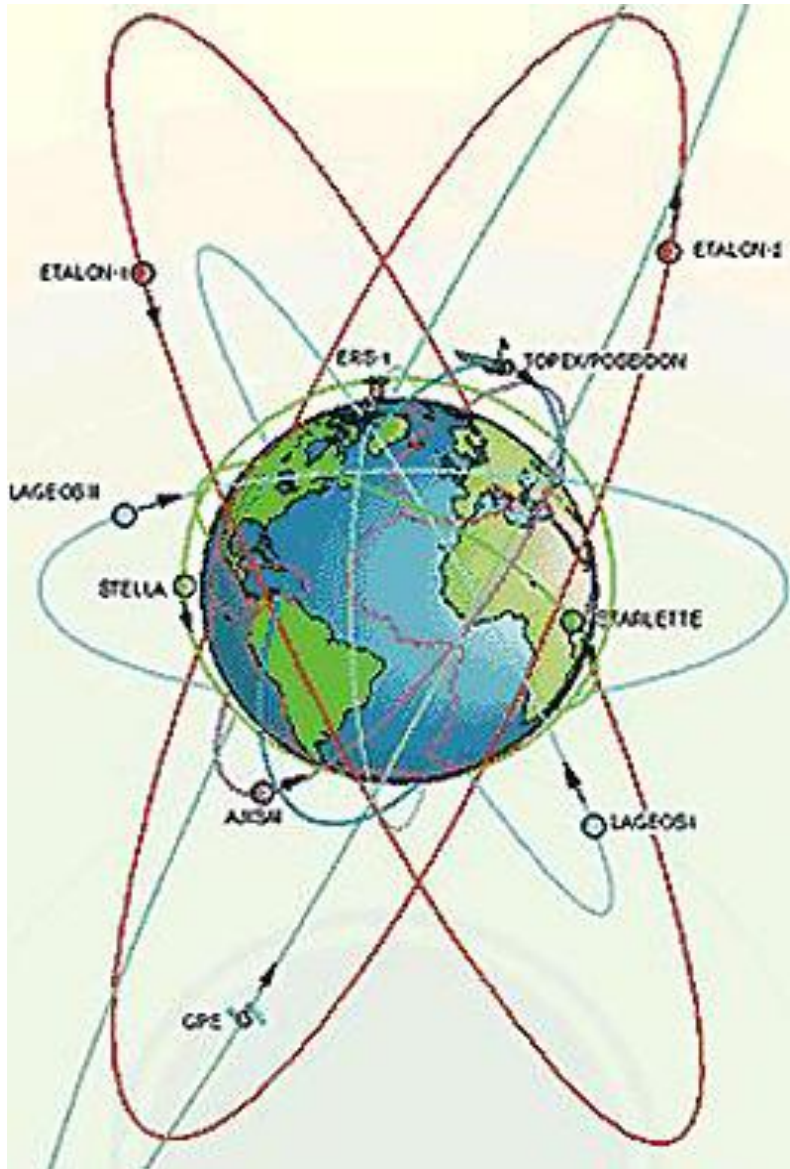
MODIS has a 1-2 day return time.

Return Time (Temporal Resolution)

- Depends on:
 - Orbital characteristics
 - Swath width
 - Ability to point the sensor

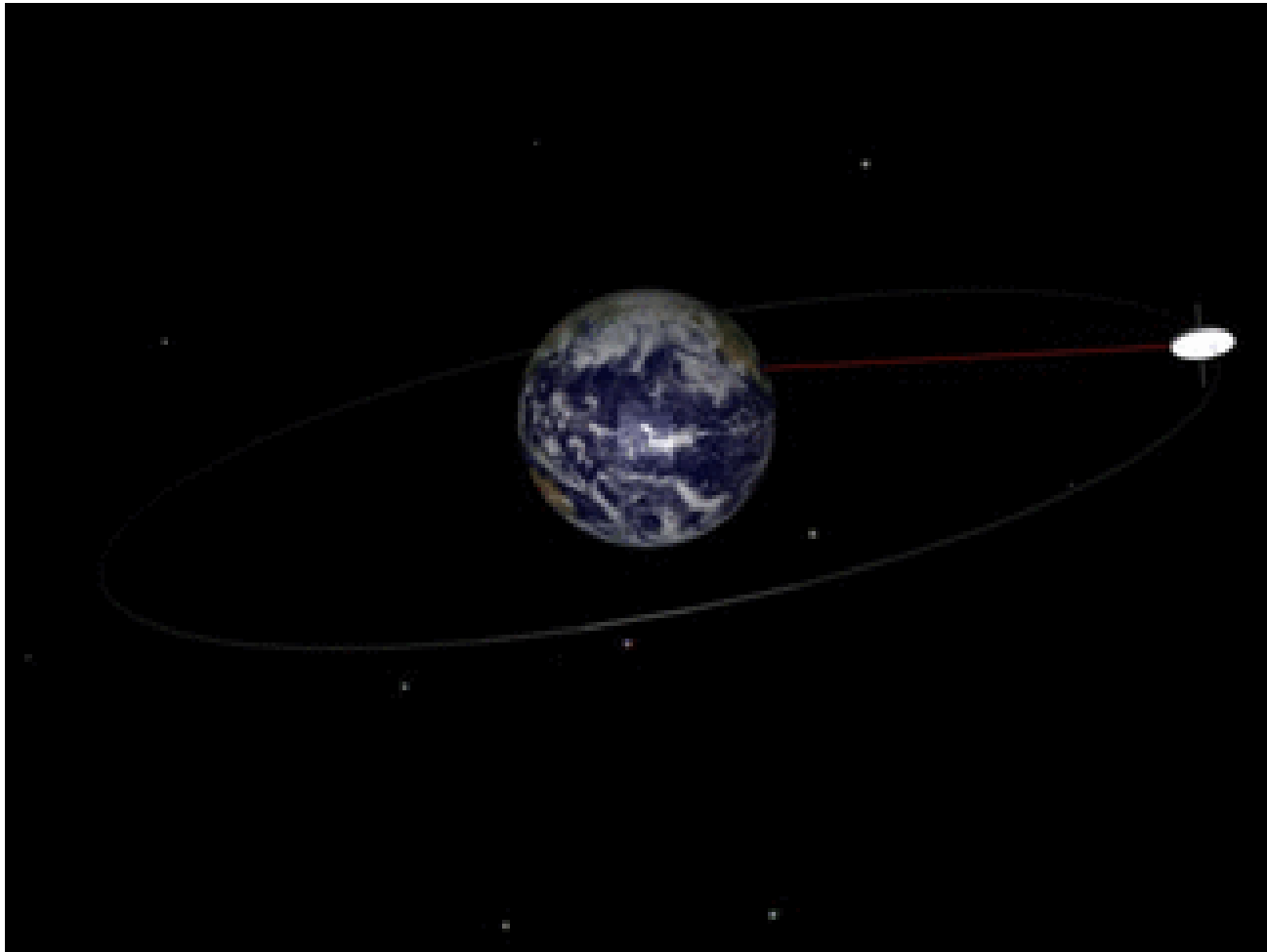
Orbital Characteristics

- Geosynchronous
- Polar
- Sun synchronous



Geosynchronous Orbits

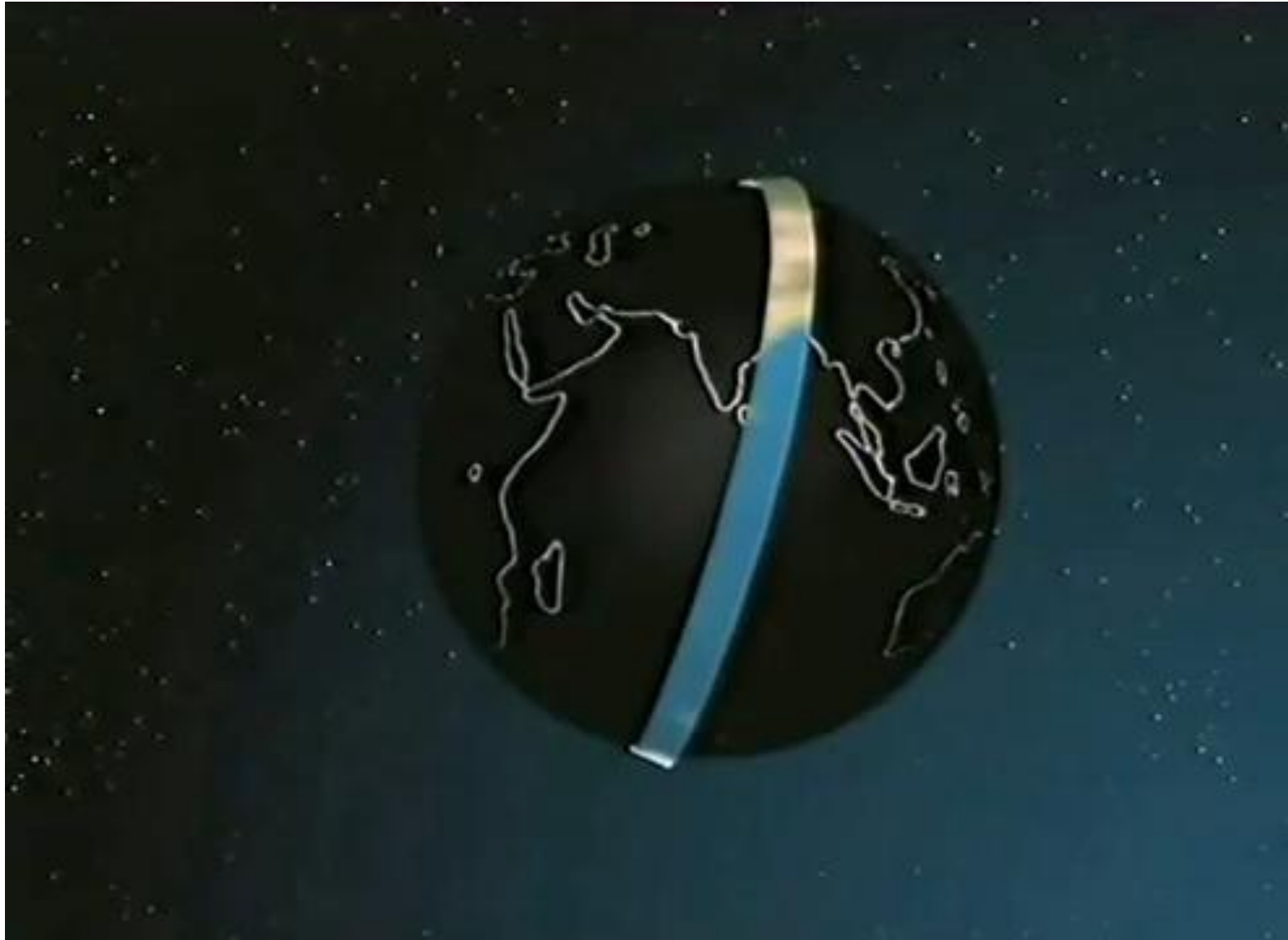
- Satellite orbits the earth at a rate that allows it to match the earth's rotation—so the satellite is always over the same place
- Narrow range of altitudes—about 35,786 km above the equator.
- Useful for communications, weather *etc.*
- Example: GOES satellite (weather)
 - Geosynchronous orbiting earth satellite



https://en.wikipedia.org/wiki/Geosynchronous_orbit#/media/File:Geosynchronous_orbit.gif

Polar/Sun Synchronous Orbits

- Pass roughly over the north and south poles
- Fly over the same place on earth at the same time of day
- Examples: Landsat, AVHRR
- Good for land remote sensing
- Return time related to spatial resolution, latitude, swath width, and orbital altitude



https://www.youtube.com/watch?v=y_jM_BxQGvE&list=PPSV

Temporal resolution = Return time:
depend on the Period of a Satellite and Swath width

$$T_0 = 2\pi(R_p + H')\sqrt{\frac{R_p + H'}{gR_p^2}}$$

T_0 = orbital period, sec.

R_p = planet radius, km (6380km for earth)

H' = orbital altitude, km

g = acceleration due to gravity (9.81m/s²)

Orbital Period for Landsat 8:

- Altitude: 705 km,
- Swath width: 185 km
- Swath overlap (or sidelap) that varies from 7% at the equator to 85% at extreme latitudes

Calculate the temporal resolution of the Landsat 8?

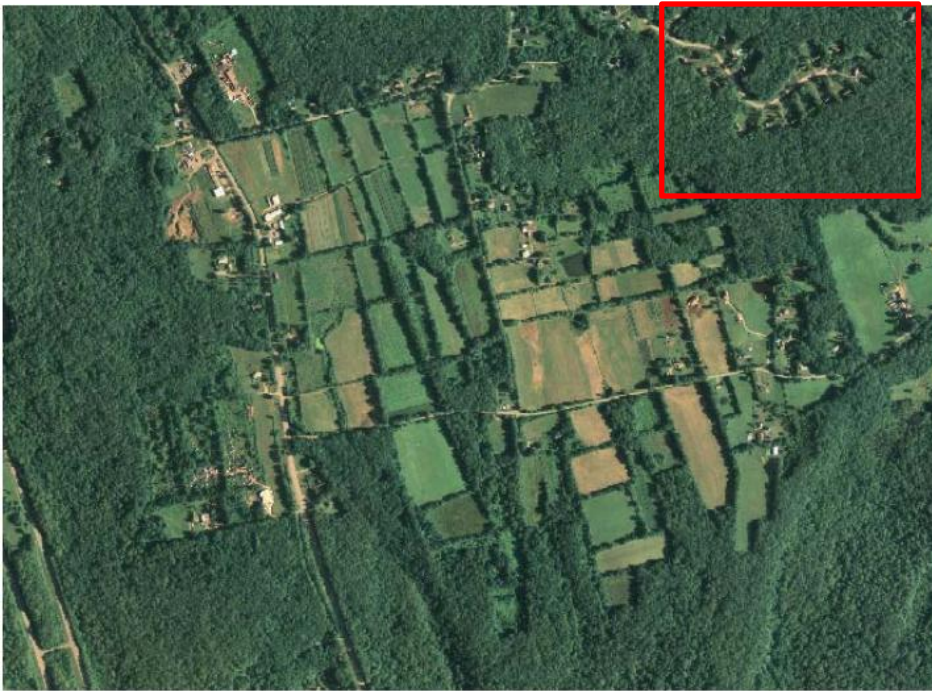
*Tips: Calculate the length of equator,
The coverage of a day,
Number of days to fully cover the equator
The repeating times of image.*

Temporal Considerations

Time of day/season image acquisition

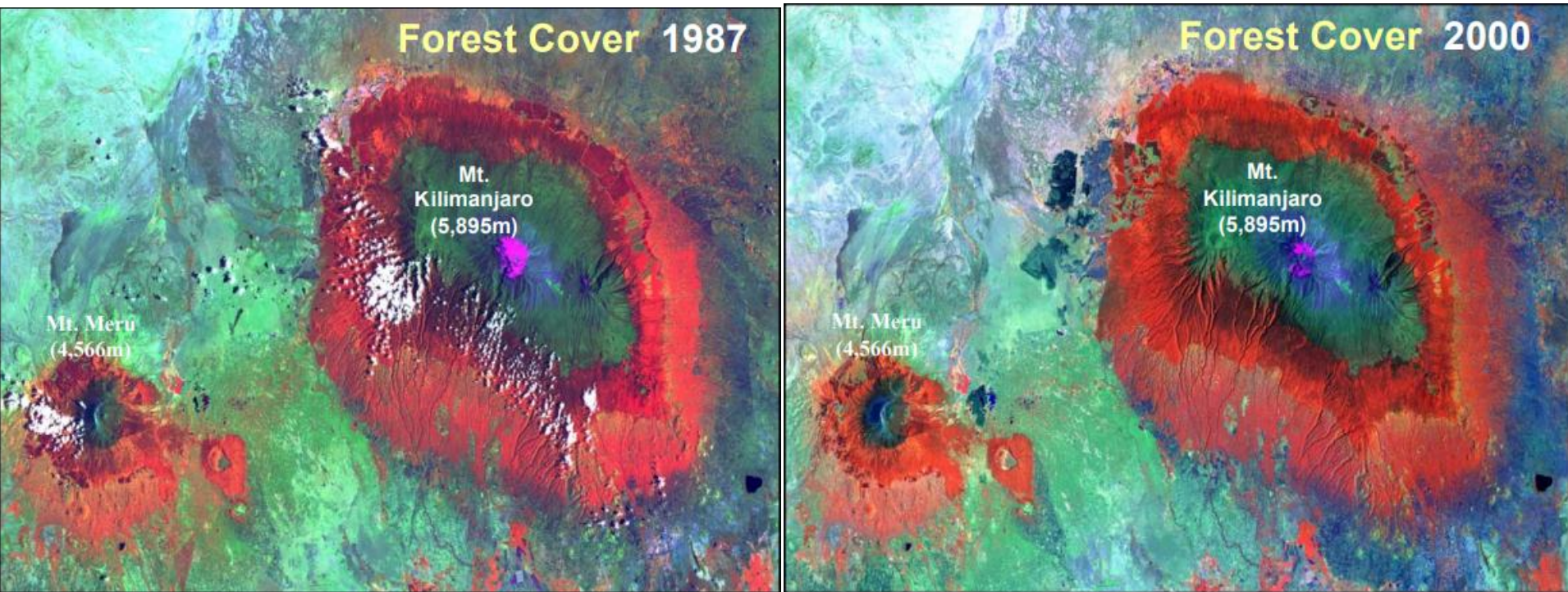
- Leaf on/leaf off
- Tidal stage
- Seasonal differences
- Shadows
- Phenological differences
- Relationship to field sampling

Seasonal Leaf-on vs. Leaf-off



Climate change?

Land cover change



Situation of forest conservation ?

Tidal stage

26/8/2016, 3.5m tide height



10/5/2014, 1.3m tide height



20/5/2010, 0.4m tide height

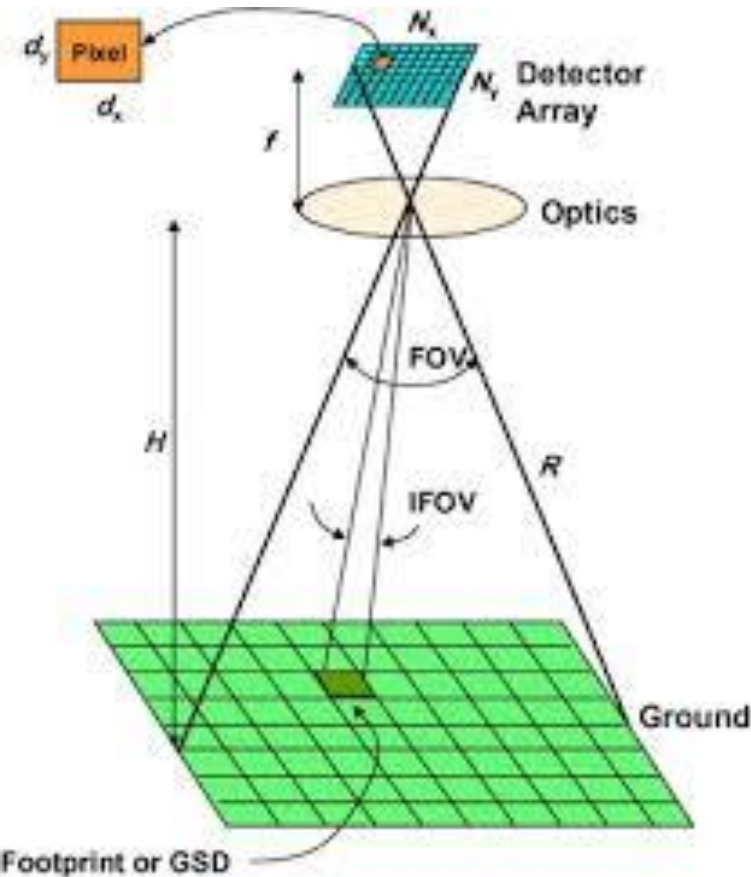


Digital elevation model of tidal flat ?

Return Time Trade Offs

- Spatial resolution
- Viewing geometry effects (off nadir)
- Clouds and other atmospheric problems
- Lack of archival repeat coverage for pointable satellites

5. Resolution TradeOffs (Geometry)



$$d/f = G/H$$

d : size of detectors

f : focal length

G : Ground distance (Spectral resolution)

H : flight height

Increase spatial resolution (reduce ground pixel size)

- Adding more/ smaller detector size
- Lower flight height
- Reduce field of view

Pixels (HxV)

512 x 512

768 x 512

1024 x 1024

1280 x 1024

1320 x 1037

1536 x 1024

2184 x 1472

2084 x 2084

3088 x 2056

4096 x 4096

Pixel Size (H x V μ m)

20.0 x 20.0

9.0 x 9.0

24.0 x 24.0

16.0 x 16.0

6.8 x 6.8

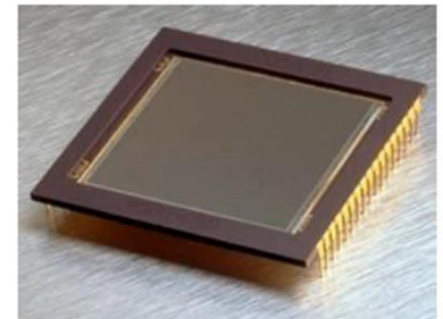
9.0 x 9.0

6.8 x 6.8

24.0 x 24.0

9.0 x 9.0

9.0 x 9.0



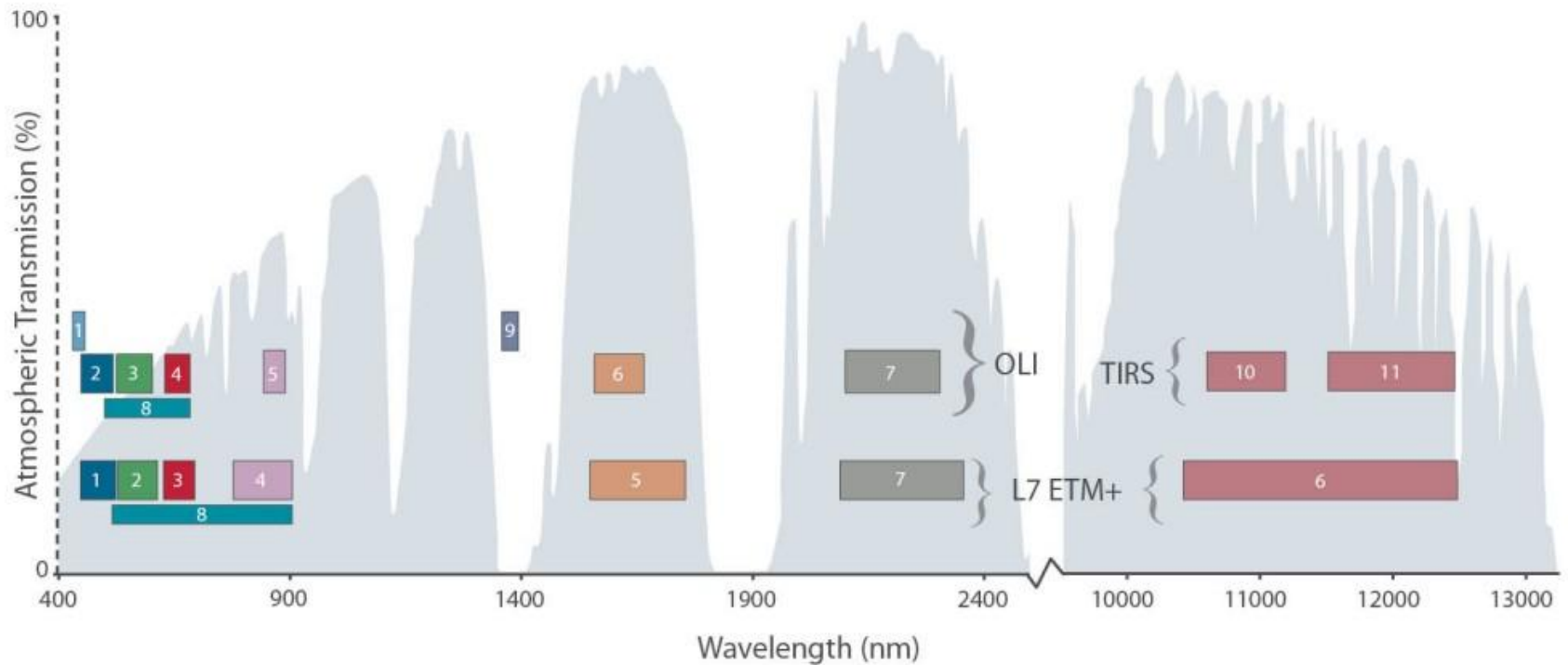
Resolution TradeOffs (Sensors)

- Signal to noise ratio (higher, better)

$$\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}}, \quad P: \text{average power } (P_{\text{signal}} \text{ high, } P_{\text{noise}} \text{ low})$$

- P_{signal} : Total energy received at sensors
 - Number of detectors: increase detectors => decrease the photons to each
 - Spectral band width: increase spectral band width => decrease the number of band
 - Dwell time (exposure time): longer camera exposure => lower flight time (temporal resolution)
- P_{noise} : Noise of sensors (sensitive, temperature..no change)

Landsat 7, 8



Characteristic of Landsat image

Landsat-7 ETM+ Bands (μm)			Landsat-8 OLI and <i>TIRS</i> Bands (μm)		
			30 m Coastal/Aerosol	0.435 - 0.451	Band 1
Band 1	30 m Blue	0.441 - 0.514	30 m Blue	0.452 - 0.512	Band 2
Band 2	30 m Green	0.519 - 0.601	30 m Green	0.533 - 0.590	Band 3
Band 3	30 m Red	0.631 - 0.692	30 m Red	0.636 - 0.673	Band 4
Band 4	30 m NIR	0.772 - 0.898	30 m NIR	0.851 - 0.879	Band 5
Band 5	30 m SWIR-1	1.547 - 1.749	30 m SWIR-1	1.566 - 1.651	Band 6
Band 6	60 m TIR	10.31 - 12.36	<i>100 m TIR-1</i>	<i>10.60 – 11.19</i>	Band 10
			<i>100 m TIR-2</i>	<i>11.50 – 12.51</i>	Band 11
Band 7	30 m SWIR-2	2.064 - 2.345	30 m SWIR-2	2.107 - 2.294	Band 7
Band 8	15 m Pan	0.515 - 0.896	15 m Pan	0.503 - 0.676	Band 8
			30 m Cirrus	1.363 - 1.384	Band 9

Increasing Signal to Noise

- Design sensor with more detectors
 - Larger detector size (more photons)
- 
- Lower spatial resolution
- Sensor record a larger range of wavelength: => lower spectral resolution

Tradeoffs: spatial, spectral, temporal resolution

- Increase spatial resolution (small pixel):

Need large bandwidth or small IFOV=> lower temporal, and spectral resolution.

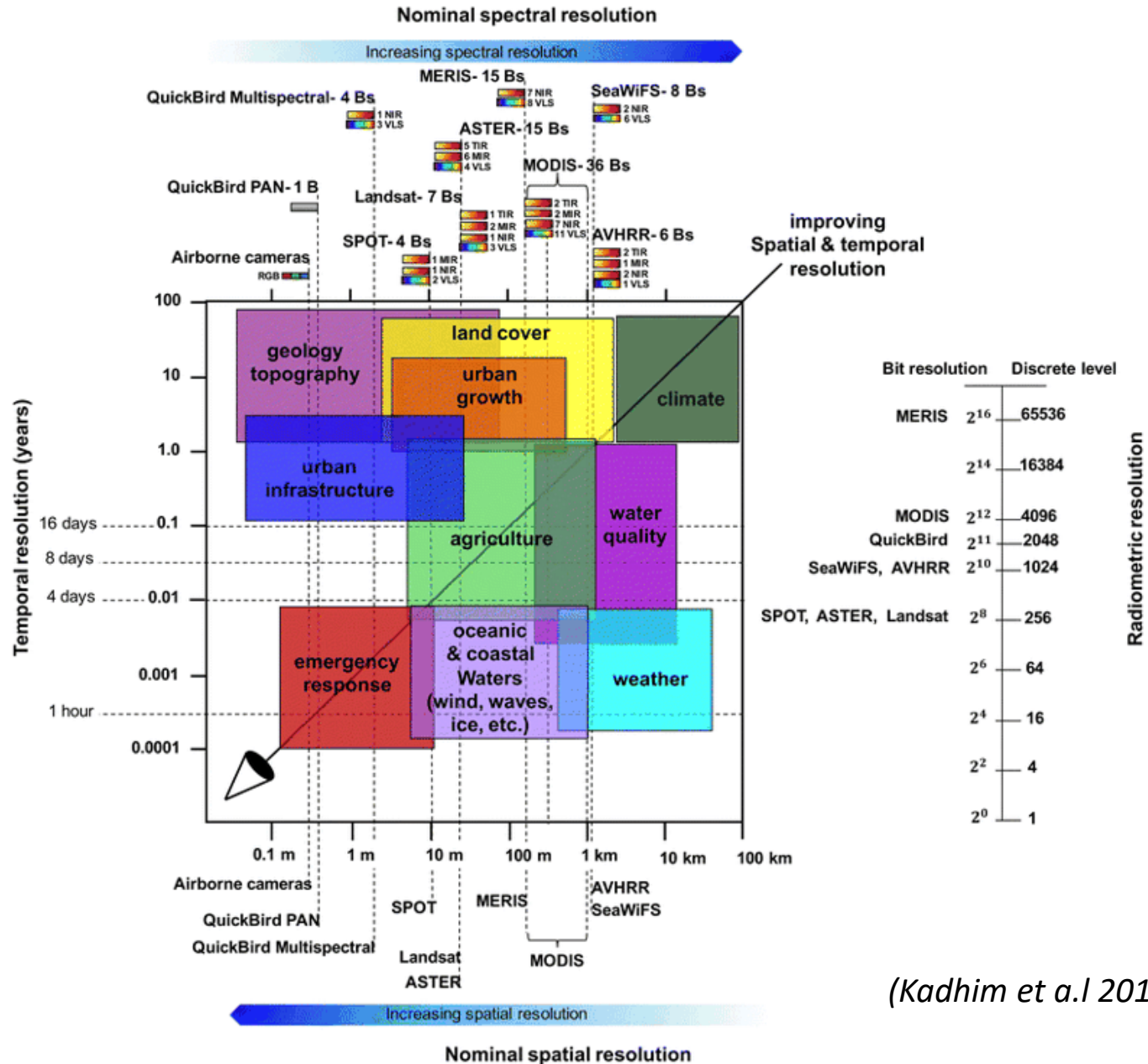
- Increase spectral resolution (more bands, narrow band):

Need larger pixel => lower temporal resolution.

- Increase temporal resolution (shorter time of revisiting):

Need larger pixels or larger IFOV => lower spatial resolution.

Resolution



(Kadhim et al 2016)

Remarks

Spatial resolution: measured in pixel size, geometric resolution or IFOV, spatial detail.

Spectral resolution: the number and width in the electromagnetic spectrum of a band

Radiometric resolution: measured in bits (binary digits), is the range of available brightness values, the depth of the image.

Temporal resolution: required time for revisiting the same area of the Earth.

Tradeoffs between resolution types: need consider to the type of solution for a specific object.

Thank you for your attention!